JACOBI COEFFICIENTS OF MEASURES ON RECIPROCAL SPECTRA: A SPECTRAL DICTIONARY AND THE NATURAL INDEX

GORDON SPEAGLE

ABSTRACT. For an increasing sequence $\Lambda = \{\lambda_m\}$ with $\lambda_m \sim m^p$ we study the recurrence coefficients a_n, b_n of the discrete measure $\sum_m \lambda_m^{-s/p}\delta_{1/\lambda_m}$; the lattice $\lambda_m = m^p$ gives the measures $\sum_k k^{-s}\delta_{k^{-p}}$ of the earlier papers of this series, for which $n^p b_n \to c(p)$. For the lattice and every $p \neq 1$ we prove an expansion of $n^p b_n/c(p)$ to order $n^{-7} + n^{-2-3p}$. It is a double series in $1/n$ and in the parameters $\zeta(-jp)n^{-jp}$ of the band factor, computed to every order by a finite residue recursion and evaluated by exact computer algebra. In the natural index $N = n + \frac{\alpha}{2} + \frac{1}{2}$, $\alpha = (s-1-p)/p$, every computed layer is even in $1/N$, and the first-order zeta terms are $\varepsilon_j N^{-1-jp}(1 + g_j N^{-2})$ with $\varepsilon_j = \binom{2j}{j} p\, c^j \zeta(-jp)$ and g_j quadratic in α, for all j. We write the full zeta layer as one Mellin–Barnes integral with kernel $\pi[\cot(\pi w p/2) - \cot(\pi w)]$, for all $p \neq 1$. On the potential theory side we solve the constrained equilibrium problem for a general decreasing node density explicitly and show that its soft edge carries every pole of the node zeta function with the weight $\pi\cot(\pi q)$. For non-lattice spectra these statements become a conjectural dictionary between the expansion of b_n and the spectral zeta function of Λ. We test it on Hurwitz lattices, Bessel zeros (where it is exact), planar drums and anharmonic oscillators; for the oscillators $-\frac{\mathrm{d}^2}{\mathrm{d}q^2} + q^N$ the predicted n^{-2} shift is the rational number $(N-1)/(6(N+2))$.

1. INTRODUCTION

1.1. **Setting.** Let $p > 0$, $s > 1$, and let $\Lambda = \{\lambda_m\}_{m\geq 1}$ be an increasing sequence of positive numbers with $\lambda_m \sim m^p$. We call Λ a *node set* and study the discrete measure on its reciprocals

$$\sigma_\Lambda := \sum_{m\geq 1} \lambda_m^{-s/p}\,\delta_{1/\lambda_m}. \tag{1.1}$$

For the *lattice* $\lambda_m = m^p$ this is the measure $\sigma_{p,s} = \sum_k k^{-s}\delta_{k^{-p}}$ of the earlier papers of this series [9, 10, 11]. Let P_n be the monic orthogonal polynomials of σ_Λ; they satisfy

$$xP_n = P_{n+1} + a_n P_n + b_n^2 P_{n-1} \qquad (n \geq 0,\ b_0 := 0,\ b_n > 0), \tag{1.2}$$

and a_n, b_n are the entries of the Jacobi matrix whose spectrum is $\{1/\lambda_m\} \cup \{0\}$. With

$$I(p) := \frac{\sqrt{\pi}}{2}\,\frac{\Gamma(\frac{1}{p} + \frac{1}{2})}{\Gamma(\frac{1}{p} + 1)}, \qquad c(p) := \frac{1}{4}\Big(\frac{\pi}{2I(p)}\Big)^p, \tag{1.3}$$

the lattice satisfies the pointwise law $n^p b_n \to c(p)$, $n^p a_n \to 2c(p)$, with rate $O(1/n)$ and an explicit $1/n$ term for every $p > 0$ [10, Theorems 1.2 and 1.4]. We write $c = c(p)$,

$$\alpha := \frac{s-1-p}{p}, \qquad y_n := \frac{n^p b_n}{c} - 1, \tag{1.4}$$

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so that $\alpha > -1$ is the Bessel order of the hard edge at 0 in [10]. The questions of this paper are: how does the expansion of y_n continue, and how does it depend on Λ? The organising object is the *node zeta function*

$$Z_\Lambda(w) := \sum_{m\ge 1} \lambda_m^{-w}, \tag{1.5}$$

continued meromorphically from $\operatorname{Re} w > 1/p$; for the lattice $Z_\Lambda(w) = \zeta(pw)$. Two quantities recur: the numbers

$$W(q) := \binom{2q}{q} c^q = \frac{(4c)^q\,\Gamma(q+\frac12)}{\sqrt{\pi}\,\Gamma(q+1)} \qquad (q \in \mathbb{C}\setminus\{-\tfrac12, -\tfrac32, \dots\}), \tag{1.6}$$

which satisfy $W(1/p) = 1$ for every p (this is the definition (1.3) of c), and the coefficients

$$\varepsilon_j := W(j)\,p\,\zeta(-jp) = \binom{2j}{j} p\,c^j \zeta(-jp) \qquad (j \ge 1), \tag{1.7}$$

so that $\varepsilon_1 = \mathsf{k}_p := 2pc\,\zeta(-p)$ is the constant of [10, Theorem 1.4(ii)].

1.2. **Results.** The first result is an expansion of the lattice coefficients far beyond the $1/n$ term. Its form is simplest in the shifted index

$$N := n + \frac{\alpha}{2} + \frac12. \tag{1.8}$$

The rational functions f_4, f_6 of (p,α) and $g_{1,4}$ are given in (5.3)–(5.5).

Theorem A (the lattice to order $n^{-7} + n^{-2-3p}$)**.** *Let $p > 0$, $p \ne 1$, $s > 1$. The recurrence coefficients of $\sigma_{p,s}$ satisfy*

$$\begin{aligned}\frac{n^p b_n}{c} = \Big(\frac{n}{N}\Big)^p \Big[1 + \frac{f_2}{N^2} + \frac{f_4}{N^4} + \frac{f_6}{N^6} + \frac{\varepsilon_1}{N^{1+p}}\Big(1 + \frac{g_1}{N^2} + \frac{g_{1,4}}{N^4}\Big) \\ + \sum_{j\ge 2} \frac{\varepsilon_j}{N^{1+jp}}\Big(1 + \frac{g_j}{N^2}\Big) + \frac{3p+1}{2p}\,\frac{\varepsilon_1^2}{N^{2+2p}}\Big] + O\big(n^{-7} + n^{-2-3p}\big),\end{aligned} \tag{1.9}$$

where the sum is over the finitely many $j \ge 2$ with $1 + jp < \min(7, 2+3p)$,

$$f_2 = \frac{p+1}{12} - \frac{(p+2)\alpha^2}{8}, \qquad g_j = A_j + B_j\alpha^2, \qquad B_j = \frac{(p+2)\big(j^2p^2 + (j+1)p + 2\big)}{8(2j-1)p},$$

$$A_j = \frac{j^2(j^2+2j-2)p^3 + 2(j+1)(2j^2-1)p^2 + (5j^2+2j-6)p + 2(j-2)}{24(2j-1)p}.$$

Theorem A is the case $K = 6$, $B = 2$ of Theorem 5.7, which gives a double expansion of $n^p b_n/c$ in $1/n$ and in the parameters $\mu_{n,j} = 2\zeta(-jp)/(jn^{jp})$ of the band factor, to every order, through a finite residue recursion. Its proof combines the third-order Riemann–Hilbert analysis of [10], extended to order K, with exact computer algebra over $\mathbb{Q}(p,\alpha)$ and, for the zeta terms, over $\mathbb{Q}(p,\alpha,j)$ with j a symbol. Three structural facts are visible in (1.9): the shift $n \mapsto N$ absorbs the whole $1/n$ term and the $(s-1)^2$ part of the n^{-2} term; every layer computed is even in $1/N$; and the first-order zeta terms are dressed by an even factor $1 + g_j N^{-2} + \cdots$ with g_j quadratic in α. Not every displayed term is a coefficient of b_n for every p: Table 5.1 lists which are.

The second result explains the zeta coefficients. The band factor of [10] enters the first-order correction through one linear functional e_n (Section 4), whose asymptotic series is $\sum_j \varepsilon_j n^{-jp}$.

Theorem B (the zeta layer as a Mellin–Barnes integral)**.** *Let $p > 0$, $p \neq 1$, and $0 < c_0 < 1$. The band-factor functional e_n of the lattice satisfies*

$$e_n = \frac{1}{2\pi i} \int_{c_0 - i\infty}^{c_0 + i\infty} p\, W(w)\, \zeta(-wp)\, n^{-wp}\, \pi \Big[\cot \frac{\pi w p}{2} - \cot(\pi w) \Big]\, \mathrm{d}w,$$

and, for every $J \geq 1$, $e_n = \sum_{j \leq J} \varepsilon_j n^{-jp} + O(n^{-(J+1)p})$. The kernel $-\pi \cot(\pi w) W(w) (4c)^{-w}$ is the Mellin transform of the truncated per-node law $(1-u)^{-1/2} \mathbf{1}_{u<1}$ (Lemma 4.1).

The third result is on the side of potential theory. In the variable $x = n^p \cdot (\text{node})$, the n nodes that carry the polynomial P_n have the continuum density $\sigma(y) = \frac{1}{p} y^{-1/p-1}$, and the zero distribution of P_n is the equilibrium measure of unit mass constrained by σ [8, 5, 9]. We solve this problem for every constraint with $y\sigma(y)$ decreasing.

Theorem C (constrained equilibrium)**.** *Let σ be a positive, locally Hölder continuous density on $(0, Y]$, $0 < Y \leq \infty$, with $y\sigma(y)$ strictly decreasing and, if $Y = \infty$, $\int_1^\infty \sigma(y) \log y \, \mathrm{d}y < \infty$. Then the soft-edge condition $\int_a^Y \sigma(y)(1 - a/y)^{-1/2} \, \mathrm{d}y = 1$ has exactly one root $a \in (0, Y)$, and the constrained equilibrium measure is $\nu \, \mathrm{d}x|_{[0,a]} + \sigma \, \mathrm{d}y|_{[a,Y]}$ with the explicit density ν of (3.2), $0 < \nu < \sigma$. For the lattice $a = 4c(p)$. For a counting function $\mathcal{N}(\lambda) \sim \sum_\rho c_\rho \lambda^\rho$, the root $a_n = 4c(1 + \eta_n) n^{-p}$ of $\int (1 - a_n \lambda)^{-1/2} \mathbf{1}_{a_n \lambda < 1} \, \mathrm{d}\mathcal{N}(\lambda) = n$ satisfies*

$$n\eta_n = p \sum_{j \geq 0} W(j) Z_{\mathcal{N}}(-j) n^{-jp} + p \sum_q R_q W(q)\, \pi \cot(\pi q)\, n^{-pq} + O(n^2 \eta_n^2),$$

the second sum over the poles $q \neq -1/p$ of $Z_{\mathcal{N}}(-w) = \int \lambda^w \, \mathrm{d}\mathcal{N}(\lambda)$, with residues R_q (Proposition 3.5).

The equilibrium statement holds up to one input from potential theory, the positivity of the logarithmic energy of compactly supported signed measures of mass zero (Lemma 3.1), which we use without proof.

1.3. **The spectral dictionary.** For the lattice, Theorems A and B identify the coefficients of b_n with values of ζ. For a general node set the same Riemann–Hilbert analysis has not been carried out, and we formulate what the theorems and the numerical evidence suggest (Section 7). In the natural index $N_\Lambda := n + \frac{\alpha}{2} - Z_\Lambda(0)$ (which is (1.8) for the lattice, $\zeta(0) = -\frac{1}{2}$), the expansion of $n^p b_n / c$ should read, at leading order in each spectral datum,

$$\Big(\frac{n}{N_\Lambda}\Big)^p \Big[1 + \frac{f_2}{N_\Lambda^2} + \cdots + \sum_{j \geq 1} \frac{W(j)\, p\, Z_\Lambda(-j)}{N_\Lambda^{1+jp}} + \sum_q \frac{p\, R_q\, W(q)\, \pi \cot(\pi q)}{N_\Lambda^{1+pq}} + \cdots \Big],$$

the second sum running over the poles q of $Z_\Lambda(-w) - \zeta(-pw)$, with w-residues R_q. The integer residues of $\pi \cot(\pi w) W(w) p Z_\Lambda(-w)$ are the zeta terms of Theorem B; its residues at the poles of Z_Λ are the pole terms of Theorem C. The dictionary is exact for the measures on reciprocal squared Bessel zeros (Proposition 5.2), it reads the index offset of the spectrum from $Z_\Lambda(0)$, and it predicts, for the eigenvalues of $-\frac{\mathrm{d}^2}{\mathrm{d}q^2} + q^N$ with N even, an n^{-2} shift equal to $(N-1)/(6(N+2))$ (Section 6.6). For drums at $p = 1$, outside the range of the theorems, the perimeter appears at order $n^{-1/2}$ and the corners at order n^{-1}, as the kernel predicts (Section 6.5). The dictionary has a limit: subleading terms of each pole, one order lower, are not given by it (Remark 7.2), and for spectra whose nodes deviate from a smooth node function by a fraction of the spacing the expansion is replaced by noise of relative size $n^{-1/2}$, whose variance appears numerically to be set by the spectral form factor (Section 6.7).

1.4. **Organisation and notation.** Section 2 restates the results of the earlier papers that are used, together with two results of the same project that extend them by one order. Section 3 proves Theorem C, Section 4 proves Theorem B, and Section 5 proves Theorem A and its general form. Section 6 collects the examples and the numerical evidence, and Section 7 the conjectures. Section 8 will relate the results to the literature. Appendix A lists the scripts.

Throughout, $p > 0$ and $c = c(p)$; α, y_n, W, ε_j, N are as in (1.4)–(1.8). The Bessel order α is that of [10], and $\mathsf{k}_p = \varepsilon_1$; the constant κ_p of [9] does not occur. We write $\binom{2q}{q} = \Gamma(2q+1)/\Gamma(q+1)^2$ for complex q. The letter N denotes the natural index, except in Section 6.6, where q^N is a potential. Constants C may change from line to line. Numbers obtained by fitting floating-point data are quoted with the spread of the fit over fit windows and nuisance exponents in parentheses; such a spread is not an error bound.

2. Setting and prior results

This section fixes the Riemann–Hilbert frame of [10] for the lattice $\sigma_{p,s}$ and restates the expansions that the later sections refine. Theorem 2.1 is proved in [10]. Theorem 2.5 and Lemmas 2.2–2.4 extend it by one order and to all band-factor parameters; their proofs are third-order versions of [10, Sections 8–10] and are not reproduced here.

2.1. The first two orders.

Theorem 2.1 ([10, Theorem 1.4])**.** *Let $s > 1$. If $p \geq 1$, then $n^p b_n/c = 1 - \frac{s-1}{2n} + O(n^{-2})$ and $n^p a_n/(2c) = 1 - \frac{s-1+p}{2n} + O(n^{-2})$. If $0 < p < 1$, both brackets carry the further term $\mathsf{k}_p n^{-1-p}$ and the remainder is $O(n^{-1-2p} + n^{-2})$.*

The $1/n$ coefficient of $n^p b_n/c$ is $d_1 := -(s-1)/2 = -p(\alpha+1)/2$. Since $(1+\delta/n)^{-p} = 1 - p\delta/n + O(n^{-2})$, it is the first term of the index shift $(n/N)^p$ with $N = n + \frac{\alpha+1}{2}$; this is the origin of (1.8).

2.2. **The frame.** We use the notation of [10, Sections 4, 8 and 10]. After the rescaling $z = n^p x$ the polynomials oscillate on the band $[0, 4c]$, with a hard edge at 0 and a soft edge at $4c$. The weight on the band is $W_0 = \mathsf{w}_n z^\alpha (4c - z) H_n(z)$, where $\mathsf{w}_n > 0$ is a constant and H_n is the *band factor*, which collects the discreteness of the nodes. With $\mathcal{R}(z) := (z(z-4c))^{1/2}$ (branch $\sim z$ at ∞), $\mathsf{T} := \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ i & -i \end{smallmatrix}\right)$ and $P_0^\infty = \mathsf{T}((z-4c)/z)^{\sigma_3/4}\mathsf{T}^{-1}$, the error matrix R of the steepest-descent analysis has jumps $I + O(1/n)$ on two small circles around 0 and $4c$ and exponentially small jumps elsewhere, and

$$n^{2p} b_n^2 = (Q_1)_{12}(Q_1)_{21}, \qquad Q_1 = P_1 + \mathsf{R}_1, \tag{2.1}$$

where P_1 and R_1 are the $1/z$ coefficients at infinity of the global parametrix and of R [10, Lemmas 4.1, 4.2 and Proposition 5.6]. The subscript "mod" denotes quantities computed for the *model weight* $z^\alpha(4c - z)$, i.e. with H_n replaced by a constant.

The band factor is $H_n = e^{-2G}$ with an explicit function G [10, Lemmas 8.1 and 10.1]. For $p > 1$, G has a jump $i\mathfrak{e}$ on $(-\infty, 0)$, with $|\mathfrak{e}(y)| \leq \pi \exp(-2\pi \sin(\pi/p) n |y|^{-1/p})$, and

$$G(z) - G(1) = \frac{1}{2\pi} \int_{-\infty}^{0} \mathfrak{e}(y) \, \frac{z-1}{(y-z)(y-1)} \, \mathrm{d}y \qquad (p > 1) \tag{2.2}$$

[10, Lemma 8.1]. For $0 < p < 1$, G jumps across the two seams $\arg z = \pm\vartheta$, $\vartheta = \pi p/2$, by $\mathfrak{j}(|z|)$, where $\mathfrak{j}(r) := -\log(1 - e^{-2\pi n r^{-1/p}})$, and

$$G(z) - G(1) = \frac{1}{2\pi i} \int_0^\infty \mathfrak{j}(r) \left[e^{i\vartheta} \mathfrak{K}(re^{i\vartheta}, z) - e^{-i\vartheta} \mathfrak{K}(re^{-i\vartheta}, z) \right] \mathrm{d}r, \qquad \mathfrak{K}(\zeta, z) := \frac{z-1}{(\zeta - z)(\zeta - 1)} \tag{2.3}$$

[10, Lemma 10.1(ii)]. Expanding $\mathfrak{K}$ in powers of z/ζ gives the moments of the jump densities:

Lemma 2.2. *Let $p \neq 1$ and $k \geq 1$. The k-th moment of the jump density, $m_k := \frac{1}{2\pi}\int_{-\infty}^{0} \mathfrak{e}(y) y^{-k-1}\,\mathrm{d}y$ for $p > 1$ and the corresponding seam integral for $p < 1$, equals $\zeta(-kp)/(kn^{kp})$. For every $K \geq 1$,*

$$\log H_n(z) - \log H_n(1) = -\sum_{k\leq K} \mu_{n,k}(z^k - 1) + O(n^{-(K+1)p}), \qquad \mu_{n,k} := \frac{2\zeta(-kp)}{kn^{kp}},$$

uniformly for $|z| \leq 4c + 2$, boundary values included, with one derivative.

For $p < 1$ this is [10, Lemma 10.1(iii)–(iv)]. For $p > 1$ the proof is the same computation on the negative axis: with $q(r) = \exp(-a_n r^{-1/p})$, $a_n = 2\pi i n e^{-i\pi/p}$, $\operatorname{Re} a_n = 2\pi n \sin(\pi/p) > 0$, one expands $\mathfrak{e} = -2\operatorname{Im}\sum_m q^m/m$, integrates termwise by [4, (5.9.1)], sums by [4, (25.2.1)] and applies the reflection formula [4, (25.4.1)]. At $p = 2$ all m_k vanish and H_n is constant, in agreement with [10, Lemma 8.1].

2.3. **Tilts and the gauge.** For $\mu = (\mu_1, \dots, \mu_K) \in \mathbb{C}^K$ put $q_\mu(x) := \exp(-\sum_k \mu_k x^k)$ and

$$Q_k(z) := \sum_{j=0}^{k-1} \binom{2j}{j} c^j z^{k-1-j}, \qquad M_\mu(z) := \sum_k \mu_k Q_k(z), \tag{2.4}$$

so that $Q_k(4c) = c^{k-1}k\binom{2k}{k}/2$. The arcsine moments $\frac{1}{\pi}\int_0^{4c} x^j (x(4c-x))^{-1/2}\,\mathrm{d}x = \binom{2j}{j}c^j$ give the Szegő function of the tilt explicitly.

Lemma 2.3 (gauge)**.** *For the weight $z^\alpha(4c - z)q_\mu(z)$, the jump data on the two circles are $\Delta_k^\mu = G_\mu \Delta_k^{\mathrm{mod}} G_\mu^{-1}$, where, with $h := \mathcal{R}M_\mu/2$ and $S := \sinh(h)/\mathcal{R}$,*

$$G_\mu(z) = P_0^\infty e^{-\mathcal{R}M_\mu\sigma_3/2}(P_0^\infty)^{-1} = \mathsf{T}\begin{pmatrix} \cosh h & -(z-4c)S \\ -zS & \cosh h \end{pmatrix}\mathsf{T}^{-1}$$

is entire in (z, μ) with $\det G_\mu = 1$. In particular Δ_k^μ is meromorphic in each disc, with Laurent coefficients entire in μ.

By Lemma 2.2, the band factor is the tilt q_{μ_n} with $\mu_n = (\mu_{n,1}, \dots, \mu_{n,K})$, up to $O(n^{-(K+1)p})$. The first-order effect of a tilt on b_n is exactly linear:

Lemma 2.4 (linear response)**.** *Let $e(\mu)$ be the $1/n$ coefficient of $\big((P_1 + \mathsf{R}_1^{(1)}(\mu)/n)_{12}(P_1 + \mathsf{R}_1^{(1)}(\mu)/n)_{21}\big)^{1/2}/c$ minus d_1, where $\mathsf{R}_1^{(1)}(\mu) = \operatorname{Res}_0 \Delta_1^\mu + \operatorname{Res}_{4c} \Delta_1^\mu$. Then, for every $\mu \in \mathbb{C}^K$,*

$$e(\mu) = cp\, M_\mu(4c) = cp \sum_k \mu_k Q_k(4c).$$

The proof expands G_μ at $4c$ and at 0. With $M_4 := M_\mu(4c)$, $G_\mu(0)$ commutes with the residue of Δ_1^{mod} at 0, $M_\mu'(4c)$ drops out, and the residue at $4c$ moves by

$$M_4 c^2 p \begin{pmatrix} M_4c - \alpha - 1 & -i(M_4c - \alpha - 2) \\ -i(M_4c - \alpha) & -(M_4c - \alpha - 1) \end{pmatrix},$$

which is quadratic in M_4 but changes $i(A_{21} - A_{12})/(2c)$, the combination that enters the $1/n$ coefficient, by exactly cpM_4. The identity $Q_k(4c) = c^{k-1}k\binom{2k}{k}/2$ follows from the recursion $a_{k+1} = 4a_k + \binom{2k}{k}$, $a_1 = 1$, satisfied by both $\sum_{j<k} 4^{k-1-j}\binom{2j}{j}$ and $k\binom{2k}{k}/2$.

2.4. **The second order.**

Theorem 2.5. *Let $p > 0$, $p \neq 1$, $s > 1$ and $K \geq 1$. Then*

$$\frac{n^p b_n}{c} = 1 + \frac{d_1}{n} + \frac{d_2}{n^2} + \sum_{j=1}^{K} \frac{\varepsilon_j}{n^{1+jp}} + O\big(n^{-3} + n^{-2-p} + n^{-1-(K+1)p}\big),$$

$$d_2 = d_2(p,s) = \frac{(p+1)(s-1)^2}{8p} - \frac{(p+2)(s-1-p)^2}{8p^2} + \frac{p+1}{12}.$$

In particular, for $0 < p < 1$ with $1/p \notin \mathbb{N}$ the n^{-2} coefficient is d_2; for $p = 1/j_0$, $j_0 \geq 2$, it is $d_2 + \varepsilon_{j_0} = d_2 - p/12$, because $W(j_0) = \binom{2j_0}{j_0} c(1/j_0)^{j_0} = 1$.

For $p > 1$ the terms with $j \geq 2$ are $O(n^{-3})$, and the statement is $n^p b_n/c = 1 + d_1/n + d_2/n^2 + \mathsf{k}_p n^{-1-p} + O(n^{-3})$. For $p = 2$, $d_2 = (s^2 + 6s - 11)/16$ and $\mathsf{k}_p = 0$. The proof of Theorem 2.5 is the second Neumann iterate of the small-norm problem, with the second asymptotic coefficients of the Bessel and Airy functions on the two circles and the band factor entering through G_{μ_n}; the $1/n^2$ term at $\mu = 0$ is an exact Laurent computation, and the zeta terms are Lemma 2.4 with $\mu_{n,j} \cdot cpQ_j(4c) = \varepsilon_j n^{-jp}$. Section 5 carries this out at every order.

3. The constrained equilibrium

At leading order the zeros of P_n, rescaled by n^p, follow the equilibrium measure of unit mass constrained by the continuum density of the nodes. For the lattice this problem was solved in [9, Theorem 1.3] in the index variable. Here we solve it for a general constraint density in the edge variable, by a Cauchy-transform argument that gives the density explicitly, and we show that the soft edge of the discrete analogue carries the poles of the node zeta function with the weight $\pi \cot(\pi q)$. Nothing in this section concerns b_n directly; the link is discussed in Section 7.

3.1. **Theorem C.** Let $0 < Y \leq \infty$ and let σ be a density on $(0, Y]$ such that

(H1) σ is positive and locally Hölder continuous on $(0, Y)$;
(H2) $y \mapsto y\sigma(y)$ is strictly decreasing on $(0, Y)$;
(H3) if $Y = \infty$, then $\int_1^\infty \sigma(y) \log y \, \mathrm{d}y < \infty$.

By (H2), $g(y) := \sqrt{y}\,\sigma(y) = y\sigma(y)/\sqrt{y}$ is strictly decreasing, and $\sigma(y) \geq y_0\sigma(y_0)/y$ for $y < y_0$, so $\int_0 \sigma = \infty$; if $Y = \infty$, (H3) gives $\int^\infty \sigma < \infty$ and hence $y\sigma(y) \to 0$. For a positive measure μ on $[0, Y]$ let $U^\mu(z) := \int \log \frac{1}{|z-x|} \, \mathrm{d}\mu(x)$ and $I(\mu) := \int U^\mu \, \mathrm{d}\mu$. The *constrained equilibrium problem* is to minimise $I(\mu)$ over positive μ on $[0, Y]$ with $\mu([0, Y]) = 1$ and $\mu \leq \sigma$. The *soft-edge condition* is

$$F(a) := \int_a^Y \sigma(y) \Big(1 - \frac{a}{y}\Big)^{-1/2} \mathrm{d}y = 1. \tag{3.1}$$

The proof of optimality uses one fact that we do not prove.

Lemma 3.1 (energy positivity)**.** *Let τ be a signed measure with compact support in $\mathbb{R}$, $\tau(\mathbb{R}) = 0$ and $\iint \big|\log|x-y|\big| \, \mathrm{d}|\tau|(x) \, \mathrm{d}|\tau|(y) < \infty$. Then $I(\tau) \geq 0$, with equality only for $\tau = 0$.*

Theorem 3.2. *Assume* (H1)–(H3).

(i) *Equation* (3.1) *has exactly one root $a \in (0, Y)$.*

(ii) *Put* $s(y) := \sigma(y)\sqrt{y/(y-a)}$ *on* (a, Y) *and*

$$\nu(x) := \frac{1}{\pi}\sqrt{\frac{a-x}{x}}\int_a^Y \frac{s(y)}{y-x}\,\mathrm{d}y \qquad (0 < x < a). \tag{3.2}$$

Then $\mu := \nu\,\mathrm{d}x|_{[0,a]} + \sigma\,\mathrm{d}y|_{[a,Y]}$ *has unit mass,* $0 < \nu < \sigma$ *on* $(0, a)$*, and* $\pi\sqrt{x(a-x)}\,\nu(x) = 1 - \int_a^Y \sigma(y)\sqrt{y(y-a)}\,(y-x)^{-1}\,\mathrm{d}y$.

(iii) *There is a constant* ℓ *with* $U^\mu = \ell$ *on* $[0, a]$ *and* $U^\mu < \ell$ *on* $(a, Y]$.

(iv) μ *is the unique solution of the constrained equilibrium problem among measures of finite energy.*

Proof. (i) Substituting $y = au$, $F(a) = \int_1^{Y/a} a\sigma(au)(1-1/u)^{-1/2}\,\mathrm{d}u$. For fixed u, $a\sigma(au) = (au\,\sigma(au))/u$ is strictly decreasing in a by (H2), and the upper limit decreases, so F is strictly decreasing. As $a \to 0$, $F(a) \geq \int_a^Y \sigma \to \infty$. If $Y < \infty$, $F(a) \to 0$ as $a \to Y$. If $Y = \infty$, split at $2a$: $\int_{2a}^\infty \leq \sqrt{2}\int_{2a}^\infty \sigma \to 0$, and, since $\sigma(y) \leq a\sigma(a)/y$ on $[a, 2a]$, $\int_a^{2a} \leq a\sigma(a)\int_1^2(1 - 1/u)^{-1/2}\,\mathrm{d}u \to 0$.

(ii) Let $S(z) := \int_a^Y s(y)(z-y)^{-1}\,\mathrm{d}y$ and $\tilde{G}(z) := ((z-a)/z)^{1/2}S(z)$ (principal branch), analytic off $[0, Y]$, and let $G(z) := \int (z-x)^{-1}\,\mathrm{d}\mu(x)$. On (a, Y), $S(z \pm i0) = \mathrm{PV}\int s(y)(z - y)^{-1}\,\mathrm{d}y \mp i\pi s(z)$ and $s = \sigma\sqrt{z/(z-a)}$, so $\tilde{G}$ jumps by $-2\pi i\sigma$; on $(0, a)$, $((z-a)/z)^{1/2}$ takes the values $\pm i\sqrt{(a-x)/x}$ and S is real, so $\tilde{G}$ jumps by $-2\pi i\nu$. These are the jumps of G. Both functions are $o(|z - z_0|^{-1})$ at $z_0 \in \{0, a, Y\}$ and tend to 0 at infinity, so $G - \tilde{G}$ is entire and vanishes at infinity, and $G = \tilde{G}$. Hence $\tilde{G}(z) \sim F(a)/z = 1/z$ at infinity, so μ has unit mass; $\operatorname{Re} G(x \pm i0) = 0$ on $(0, a)$; and $\nu > 0$ because $s > 0$. With $R(y) := \sqrt{y(y-a)}$ one has $R\sigma = s\,(y - a)$, which gives $\tilde{G} = R(z)^{-1}\big[F(a) + \int \sigma R\,(z-y)^{-1}\,\mathrm{d}y\big]$; the difference of the two expressions for ν is $(F(a) - 1)/(\pi\sqrt{x(a-x)})$, which vanishes by (3.1). For the upper bound write $s(y) = g(y)(y-a)^{-1/2}$ and use $\int_a^\infty (y-a)^{-1/2}(y-x)^{-1}\,\mathrm{d}y = \pi(a-x)^{-1/2}$:

$$\nu(x) < \frac{1}{\pi}\sqrt{\frac{a-x}{x}}\,g(a)\,\frac{\pi}{\sqrt{a-x}} = \frac{g(a)}{\sqrt{x}} \leq \frac{g(x)}{\sqrt{x}} = \sigma(x) \qquad (0 < x < a).$$

(iii) On $(0, a)$, U^μ is constant because $\operatorname{Re} G = 0$ there. For $z \in (a, Y)$, $\varphi := U^\mu - \ell$ satisfies $\varphi(a) = 0$ and $\varphi'(z) = -\operatorname{Re} G(z \pm i0) = -((z-a)/z)^{1/2}H(z)$ with $H(z) := \mathrm{PV}\int_a^Y s(y)(z - y)^{-1}\,\mathrm{d}y$, which exists since σ is locally Hölder. Writing $s = g \cdot (y-a)^{-1/2}$,

$$H(z) = \int_a^Y \frac{(g(y) - g(z))(y-a)^{-1/2}}{z-y}\,\mathrm{d}y + g(z)\,\mathrm{PV}\int_a^Y \frac{(y-a)^{-1/2}}{z-y}\,\mathrm{d}y.$$

The first integrand is positive since g is decreasing. The principal value is 0 if $Y = \infty$, and $(z-a)^{-1/2}\log\frac{\sqrt{Y-a}+\sqrt{z-a}}{\sqrt{Y-a}-\sqrt{z-a}} > 0$ if $Y < \infty$. Hence $H > 0$, $\varphi' < 0$ and $\varphi < 0$ on $(a, Y]$.

(iv) Let $\mu' \leq \sigma$ have unit mass and finite energy, and $\tau := \mu' - \mu$. Then $I(\mu') - I(\mu) = I(\tau) + 2\int (U^\mu - \ell)\,\mathrm{d}\tau$. The integrand vanishes on $[0, a]$; on $(a, Y]$, $\tau \leq 0$ and $U^\mu - \ell < 0$, so the integral is ≥ 0. For $I(\tau)$, map $x \mapsto x^* := 1/(1+x)$, which sends $[0, Y]$ into $(0, 1]$; since $\log\frac{1}{|x^*-y^*|} = \log\frac{1}{|x-y|} + \log(1+x) + \log(1+y)$ and $\tau(\mathbb{R}) = 0$, the extra terms integrate to $2\tau(\mathbb{R})\int \log(1+x)\,\mathrm{d}\tau = 0$ (finite by (H3), since $\mu, \mu' \leq \sigma$). So $I(\tau) = I(\tau^*) \geq 0$ by Lemma 3.1, with equality only if $\tau = 0$. □

Remark 3.3. (a) The proof of (ii)–(iii) uses only that g is decreasing; (H2) enters through the monotonicity of F in (i). So (H2) may be replaced by "$\sqrt{y}\,\sigma(y)$ strictly decreasing and $\int_0^Y \sigma > 1$"; uniqueness of the root then follows from (iv), since a is the left end of the saturated

set of the unique minimiser. (b) Without (3.1), ν acquires a multiple of $(a-x)^{-1/2}$, which violates $\nu \geq 0$ or $\nu \leq \sigma$. Under (3.1), $\nu(a^-) = g(a)/\sqrt{a} = \sigma(a)$, so μ is continuous at a, and $\sigma - \nu$ vanishes like a constant times $\sqrt{a-x}$. (c) In node language (3.1) is the statement that $\frac{1}{n}\sum_{u_k<1}(1-u_k)^{-1/2} \approx 1$, with $u_k = a\lambda_k/n^p$: the edge is fixed by a sum of the per-node quantity $(1-u)^{-1/2}$ over the saturated nodes.

Corollary 3.4 (the lattice)**.** *For $\sigma(y) = \frac{1}{p}y^{-1/p-1}$ on $(0,\infty)$ the root of (3.1) is $a = 4c(p)$.*

Proof. $y\sigma(y) = \frac{1}{p}y^{-1/p}$ is strictly decreasing and (H3) holds. Equation (3.1) reads $a^{-1/p}\frac{1}{p}B(\frac{1}{p},\frac{1}{2}) = 1$, i.e. $a^{1/p} = \sqrt{\pi}\,\Gamma(\frac{1}{p}+1)/\Gamma(\frac{1}{p}+\frac{1}{2}) = \pi/(2I(p))$, and $a = 4c(p)$ by (1.3). □

The density $\sigma(y) = \frac{1}{p}y^{-1/p-1}$ is the image of Lebesgue measure on the index variable u under $y = u^{-p}$, so Corollary 3.4 concerns the problem of [9, Theorem 1.3]; its edge $a = 4c$ is the right end of the interval on which the potential of the measure μ^* there is constant [9, Theorem 1.3(ii)]. The derivation above does not use the Riemann–Hilbert analysis of [10].

3.2. **The soft edge carries the poles.** Let $L(u) := (1-u)^{-1/2}\mathbf{1}_{u<1}$; its Mellin transform $\widetilde{L}(w) := \int_0^\infty u^{-w-1}L(u)\,\mathrm{d}u$ is computed in Lemma 4.1: $\widetilde{L}(w) = B(-w,\frac{1}{2}) = -\pi\cot(\pi w)\,\Gamma(w+\frac{1}{2})/(\sqrt{\pi}\,\Gamma(w+1))$, so that

$$-(4c)^q\,\widetilde{L}(q) = W(q)\,\pi\cot(\pi q). \tag{3.3}$$

At $q \in -\frac{1}{2} - \mathbb{N}_0$ the right side is a limit of the form $0\cdot\infty$ and is defined by the left side; for instance $\widetilde{L}(-\frac{1}{2}) = \pi$.

Let $\mathcal{N}$ be a C^1 counting function on $[\lambda_0,\infty)$, $\lambda_0 > 0$, with $\mathcal{N}' > 0$, and suppose that for exponents $\rho_0 = 1/p > \rho_1 > \rho_2 > \cdots$ and constants c_ρ with $c_{1/p} = 1$,

$$\mathcal{N}'(\lambda) = \sum_{k<K} \rho_k c_{\rho_k}\lambda^{\rho_k-1} + O(\lambda^{\rho_K-1}) \qquad (\lambda\to\infty) \tag{3.4}$$

for every K (if the list is finite, the remainder is $O(\lambda^{-M})$ for every M). Then $Z_{\mathcal{N}}(s) := \int_{\lambda_0}^\infty \lambda^{-s}\,\mathrm{d}\mathcal{N}(\lambda)$ continues meromorphically to $\mathbb{C}$, with simple poles at $s = \rho$ of residue ρc_ρ. In the variable $w = -s$, $w \mapsto Z_{\mathcal{N}}(-w)$ has poles at $q = -\rho$ with residues $R_q = -\rho c_\rho$; the *Weyl pole* is $q = -1/p$, $R_{-1/p} = -1/p$. We assume that $\rho c_\rho = 0$ whenever $\rho \in \{-1,-2,\dots\}$, i.e. no pole of $Z_{\mathcal{N}}(-w)$ sits at a positive integer.

Proposition 3.5. *Under these assumptions, as $A \to 0^+$,*

$$\int_{\lambda_0}^\infty L(A\lambda)\,\mathrm{d}\mathcal{N}(\lambda) \sim -\sum_q R_q\widetilde{L}(q)A^q + \sum_{j\geq 0}\binom{2j}{j}4^{-j}Z_{\mathcal{N}}(-j)A^j, \tag{3.5}$$

with remainder $O(A^{-\rho_K})$ after the terms with $q < -\rho_K$ and $j < -\rho_K$. Consequently the root of $\int L(A\lambda)\,\mathrm{d}\mathcal{N}(\lambda) = n$, written $A = 4c(1+\eta)n^{-p}$, satisfies

$$\eta = \frac{p}{n}\Big[\sum_{j\geq 0} W(j)Z_{\mathcal{N}}(-j)n^{-jp} + \sum_{q\neq -1/p} R_q W(q)\,\pi\cot(\pi q)\,n^{-pq}\Big] + O(n\eta^2), \tag{3.6}$$

and the higher orders follow by reversion of the series.

Proof. For one power, $\mathrm{d}\mathcal{N} = \rho c_\rho\lambda^{\rho-1}\,\mathrm{d}\lambda$, the substitution $u = A\lambda$ and the Taylor expansion of $(1-u)^{-1/2}$ on $[0, A\lambda_0]$ give

$$\int_{\lambda_0}^{1/A}(1-A\lambda)^{-1/2}\rho c_\rho\lambda^{\rho-1}\,\mathrm{d}\lambda = \rho c_\rho\Big[A^{-\rho}B(\rho,\tfrac{1}{2}) - \sum_{j\geq 0}\binom{2j}{j}4^{-j}A^j\frac{\lambda_0^{\rho+j}}{\rho+j}\Big],$$

first for $\rho > 0$ and then for all $\rho \notin -\mathbb{N}$ by analytic continuation in ρ. The first term is $-R_q\widetilde{L}(q)A^q$ with $q = -\rho$. For the remainder r_K of (3.4), split at $\lambda = 1/(2A)$: on $[\lambda_0, 1/(2A)]$ expand $(1 - A\lambda)^{-1/2}$ in powers of $A\lambda$, and on $[1/(2A), 1/A]$ use $|r_K| \le C\lambda^{\rho_K - 1}$; both give the integer terms up to $O(A^{-\rho_K})$. The coefficient of A^j collects $-\rho c_\rho \lambda_0^{\rho+j}/(\rho + j)$ over the powers and $\int \lambda^j r_K \, \mathrm{d}\lambda$, i.e. the continued value $Z_{\mathcal{N}}(-j)$. This proves (3.5). The Weyl pole contributes $\frac{1}{p}B(\frac{1}{p}, \frac{1}{2})A^{-1/p} = n(1 + \eta)^{-1/p}$ by Corollary 3.4, so the equation is $n(1 + \eta)^{-1/p} + (\text{rest}) = n$; by (3.3) and $\binom{2j}{j}4^{-j}(4c)^j = W(j)$, the rest at $\eta = 0$ is the bracket of (3.6), and the implicit function theorem gives (3.6). □

The termwise proof avoids shifting the contour of the Mellin–Barnes form

$$\frac{1}{2\pi i}\int \widetilde{L}(w)Z_{\mathcal{N}}(-w)A^w \, \mathrm{d}w$$

of (3.5), which would need bounds on $Z_{\mathcal{N}}$ on vertical lines. The numbers $Z_{\mathcal{N}}(-j)$ depend on the cutoff λ_0; they are not the values $Z_\Lambda(-j)$ of a node set, which enter through the band factor (Section 4). What (3.6) contributes for a node set is the pole part: the weight $W(q)\pi\cot(\pi q)$ of each pole of $Z(-w)$ other than the Weyl pole.

Remark 3.6 (exact root for a drum-type count). For $p = 1$ ($c = \frac{1}{2}$) and $\mathrm{d}\mathcal{N} = (1 - \frac{\beta}{2}\lambda^{-1/2})\,\mathrm{d}\lambda$ on $(0, \infty)$, the condition is $2/A - \frac{\pi\beta}{2}A^{-1/2} = n$. With $A = 2(1 + \eta)/n$ and $v := (1 + \eta)^{-1/2}$ it reads $v^2 - \epsilon v - 1 = 0$, $\epsilon := \pi\beta/(2\sqrt{2n})$, so $1 + \eta = v^{-2} = 1 - \epsilon + \epsilon^2/2 + O(\epsilon^3)$ and

$$n\eta = -\frac{\pi\beta}{2\sqrt{2}}\, n^{1/2} + \frac{\pi^2}{16}\beta^2 + O(n^{-1/2}).$$

The first term is the pole term of (3.6) at $q = -\frac{1}{2}$, $R_q = \beta/2$, with $W(q)\pi\cot(\pi q) = -\pi(4c)^{-1/2}$; the second is quadratic in the pole residue and is produced by the reversion. Both are compared with b_n in Section 6.5.

Remark 3.7 (numerical checks). For the lattice at $p = 0.6$ and $p = 1.5$, for the drum density of Remark 3.6 cut at the first node, and for a fourth density, the root, the two forms of ν, the mass, $0 < \nu/\sigma < 1$ and the outer inequality of Theorem 3.2 were evaluated by quadrature; the root agrees with $4c$ to 10^{-12} in the lattice cases, and the two forms of ν to 10^{-21}, which tests the root and the quadrature rather than (ii). The first-order pole term of (3.6) was fitted from the root over $n = 100$–6400 for $\lambda_k = (k + \gamma k^{-m})^p$ at $(p, m) = (1.5, \frac{1}{2}), (0.75, \frac{1}{4}), (1.25, \frac{1}{2}), (1.8, \frac{3}{4})$; it agrees with $pR_qW(q)\pi\cot(\pi q)$ to between 10^{-9} and 10^{-6} (scripts `m1_theoremQ`, `m1_edge`).

4. The kernel

By Lemma 2.4 and Lemma 2.2, the zeta terms of Theorem 2.5 are the values at $\mu = \mu_n$ of the linear functional $cpM_\mu(4c)$, truncated to K terms. This section evaluates the same functional on the band factor itself, without truncation, and writes it as one Mellin–Barnes integral (Theorem 4.2). Its kernel contains $-\pi\cot(\pi w)$, which is also the Mellin transform of the truncated per-node law of Section 3 (Lemma 4.1). The proofs are for the lattice; Section 7 discusses other node sets.

4.1. The truncated per-node law.

Lemma 4.1. *Let* $L(u) := (1 - u)^{-1/2}\mathbf{1}_{u<1}$. *For* $\operatorname{Re} w < 0$,

$$\widetilde{L}(w) := \int_0^\infty u^{-w-1}L(u)\,\mathrm{d}u = B\big(-w, \tfrac{1}{2}\big) = -\pi\cot(\pi w)\,\frac{\Gamma(w + \frac{1}{2})}{\sqrt{\pi}\,\Gamma(w + 1)}.$$

$\widetilde{L}$ continues meromorphically to $\mathbb{C}$; its poles in $\operatorname{Re} w \geq 0$ are simple, at $w = j \in \mathbb{N}_0$, with $-\operatorname{Res}_{w=j} \widetilde{L} = \binom{2j}{j} 4^{-j}$, the Taylor coefficients of L at 0. For $a = 4c$, $(a/n^p)^w \Gamma(w + \frac{1}{2})/(\sqrt{\pi}\Gamma(w+1)) = W(w) n^{-pw}$.

Proof. The integral is $B(-w, \frac{1}{2}) = \Gamma(-w)\sqrt{\pi}/\Gamma(\frac{1}{2} - w)$ by [4, (5.12.1)]. By the reflection formula [4, (5.5.3)], $\Gamma(\frac{1}{2} - w)\Gamma(\frac{1}{2} + w) = \pi/\cos(\pi w)$ and $\Gamma(-w)\Gamma(1+w) = -\pi/\sin(\pi w)$, so $\sqrt{\pi}\,\Gamma(-w)/\Gamma(\frac{1}{2} - w) = \Gamma(-w)\Gamma(w + \frac{1}{2})\cos(\pi w)/\sqrt{\pi} = -\pi \cot(\pi w)\Gamma(w + \frac{1}{2})/(\sqrt{\pi}\,\Gamma(w+1))$. The residue of $-\pi\cot(\pi w)$ at j is -1, and $\Gamma(j + \frac{1}{2})/(\sqrt{\pi}\, j!) = \binom{2j}{j} 4^{-j}$. □

The two analytic continuations of $(1-u)^{-1/2}$ to $u > 1$ are $\mp i|1-u|^{-1/2}$, and their Mellin transforms carry $\Gamma(-w)\Gamma(1+w)e^{\pm i\pi w} = -\pi e^{\pm i\pi w}/\sin(\pi w)$. Their average is L, and the average of the two transforms is $-\pi\cot(\pi w)$: the cotangent is the signature of truncating the per-node law at the edge $u = 1$.

4.2. **The band-factor functional.** For a function f analytic near the jump contour of G put, with the notation of (2.2)–(2.3),

$$\mathcal{M}_n[f] := \frac{1}{2\pi} \int_{-\infty}^{0} \mathfrak{e}(y) f(y)\, \mathrm{d}y \qquad (p > 1),$$

$$\mathcal{M}_n[f] := \frac{1}{\pi} \int_{0}^{\infty} \mathfrak{j}(r)\, \operatorname{Im}\big[e^{i\vartheta} f(re^{i\vartheta})\big]\, \mathrm{d}r \qquad (0 < p < 1),$$

so that $\mathcal{M}_n[y^{-k-1}] = m_k = \zeta(-kp)/(kn^{kp})$ by Lemma 2.2 (for $0 < p < 1$ the integrand is real and $\mathcal{M}_n$ is the seam integral of [10, Lemma 10.1(iii)]). Since $\sum_{j\geq 0} \binom{2j}{j} c^j t^j = (1-4ct)^{-1/2}$ and $\sum_l (4c)^l t^l = (1-4ct)^{-1}$,

$$\sum_{k\geq 1} Q_k(4c)\, y^{-k-1} = k_c(y) := y^{-2}\Big(1 - \frac{4c}{y}\Big)^{-3/2} \qquad (|y| > 4c). \tag{4.1}$$

The *band-factor functional* is

$$e_n := 2cp\, \mathcal{M}_n[k_c], \tag{4.2}$$

that is, $e_n = \frac{cp}{\pi} \int_0^\infty \mathfrak{e}(-r)\, r^{-1/2}(r+4c)^{-3/2}\, \mathrm{d}r$ for $p > 1$ and $e_n = \frac{2cp}{\pi} \int_0^\infty \mathfrak{j}(r) \operatorname{Im}[e^{i\vartheta} k_c(re^{i\vartheta})]\, \mathrm{d}r$ for $0 < p < 1$. On the seams $\operatorname{Im}(1 - 4c/\zeta) = (4c/r)\sin\vartheta > 0$, so the principal branch in k_c is never crossed. By Lemma 2.4 and Lemma 2.2, $e(\mu_n) = cp \sum_{k\leq K} \mu_{n,k} Q_k(4c) = 2cp \sum_{k\leq K} m_k Q_k(4c)$ is the truncation of e_n obtained by expanding k_c in (4.1).

Theorem 4.2. *Let $p > 0$, $p \neq 1$, $s > 1$, and $0 < c_0 < 1$.*

(i) $e_n = 2cp \sum_{k\leq K} m_k Q_k(4c) + O(n^{-(K+1)p})$ *for every K.*

(ii) $e_n = \dfrac{1}{2\pi i} \displaystyle\int_{c_0 - i\infty}^{c_0 + i\infty} p\, W(w)\, \zeta(-wp)\, n^{-wp}\, \pi \Big[\cot\frac{\pi w p}{2} - \cot(\pi w)\Big]\, \mathrm{d}w.$

(iii) *For every $J \geq 1$, $e_n = \sum_{j\leq J} \varepsilon_j n^{-jp} + O(n^{-(J+1)p})$.*

Part (iii) follows from (i) and $2cpm_jQ_j(4c) = \varepsilon_j n^{-jp}$; the content of the theorem is the exact representation (ii) and its kernel.

Proof. (i) Split the integral at $|y| = 2(4c+2)$. On the far part, expand k_c by (4.1) to K terms with remainder $O(|y|^{-K-2})$; the near part, and the near parts of the moments, are $O(e^{-n/C})$ because the jump densities are [10, Lemmas 8.1(i) and 10.1]. The remainder is bounded as in Lemma 2.2.

(ii), $p > 1$. For complex w with $\operatorname{Re} w > 0$, the computation of Lemma 2.2 with k replaced by w (the Fubini bound depends only on $\operatorname{Re} w$) gives

$$I(w) := \int_0^\infty \mathfrak{e}(-r) r^{-w-1}\,\mathrm{d}r = -2p\,\Gamma(wp)\zeta(1+wp)(2\pi n)^{-wp}\sin\Big(\pi w - \frac{\pi wp}{2}\Big).$$

The strip is set by $\mathfrak{e}(-r) \to \pi(1-2/p)$ as $r \to \infty$ and by the exponential decay of $\mathfrak{e}(-r)$ as $r \to 0$. By [4, (25.4.1)] at $1+wp$, $p\Gamma(wp)\zeta(1+wp)(2\pi)^{-wp} = -\pi\zeta(-wp)/(w\sin(\pi wp/2))$, so

$$I(w) = \frac{2\pi}{w}\,\zeta(-wp)\,n^{-wp}\Big[\sin(\pi w)\cot\frac{\pi wp}{2} - \cos(\pi w)\Big].$$

Next, $r^{-1/2}(r+4c)^{-3/2} = r^{-2}(1+4c/r)^{-3/2}$, and the Mellin–Barnes representation

$$(1+x)^{-3/2} = \frac{1}{2\pi i}\int_{(c_1)} \frac{\Gamma(-s)\Gamma(\frac32+s)}{\Gamma(\frac32)}\,x^s\,\mathrm{d}s \qquad (x>0,\ -\tfrac32 < c_1 < 0) \tag{4.3}$$

is the Mellin inversion of the Beta integral [4, (5.12.3)], whose transform is absolutely integrable on the line. Put $w = s+1$, so that $\operatorname{Re} w = c_0 \in (0,1)$, and exchange the integrals (absolute convergence). Using $\Gamma(1-w)/w = \pi/(\sin(\pi w)\Gamma(1+w))$ and $\Gamma(\frac32) = \sqrt{\pi}/2$, the factors collect into $pW(w)\zeta(-wp)n^{-wp}\pi[\sin(\pi w)\cot(\pi wp/2) - \cos(\pi w)]/\sin(\pi w)$, which is the stated integrand.

(ii), $0 < p < 1$. The Mellin transform of the seam density is $\int_0^\infty \mathfrak{j}(r) r^{-w-1}\,\mathrm{d}r = p(2\pi n)^{-wp}\Gamma(wp)\zeta(1+wp)$ for $\operatorname{Re} w > 0$, by the computation of [10, Lemma 10.1(iii)] with complex w; the strip comes from $\mathfrak{j}(r) \sim \frac{1}{p}\log r$ at infinity and the super-exponential decay at 0. Expand $(1+v)^{-3/2}$ with $v = -4ce^{-i\vartheta}/r$ by (4.3) (now $\arg v = \pi - \vartheta \in (0,\pi)$, where the representation continues). With $w = s+1$ the angular factor is $\operatorname{Im}[e^{i\vartheta}e^{-2i\vartheta}e^{-i\vartheta s}e^{i\pi s}] = -\sin((\pi-\vartheta)w)$, and with $\vartheta = \pi p/2$ and the same two identities the integrand becomes $pW(w)\zeta(-wp)n^{-wp}\pi\sin(\pi w - \frac{\pi wp}{2})/(\sin(\pi w)\sin\frac{\pi wp}{2})$. Finally $\sin(\pi w - \frac{\pi wp}{2})/(\sin(\pi w)\sin\frac{\pi wp}{2}) = \cot\frac{\pi wp}{2} - \cot(\pi w)$.

(iii) On vertical lines $|\cot(\pi wp/2) - \cot(\pi w)| \le Ce^{-\min(p,2)\pi|\operatorname{Im} w|}$, while $\zeta(-wp)$ and $W(w)$ grow at most polynomially, so the contour can be moved to the right. The poles of $-\pi\cot(\pi w)$ at $w = j \ge 1$ give $-\operatorname{Res} = pW(j)\zeta(-jp)n^{-jp} = \varepsilon_j n^{-jp}$. The poles of $\cot(\pi wp/2)$ at $w = 2l/p$ lie on the zeros $\zeta(-2l) = 0$ and are removable; where $2l/p = j$ is an integer the product is regular and $\varepsilon_j = 0$. □

Remark 4.3 (reading of Theorem 4.2). (a) The second kernel $\pi\cot(\pi wp/2)$ comes only from writing $I(w)$ with $\zeta(-wp)$: in the form $-2p\Gamma(wp)\zeta(1+wp)(2\pi n)^{-wp}\sin(\pi w - \pi wp/2)$ there are no poles at $w = 2l/p$. So the band factor contributes exactly the integer residues of $-\pi\cot(\pi w)$. (b) The pole of $\zeta(-wp)$ at $w = -1/p$ (the Weyl pole) and the point $w = 0$ lie to the left of the strip. The $j = 0$ term $pW(0)Z(0) = p\zeta(0)$, which is part of d_1, does not come from e_n but from the index normalisation of the frame. (c) e_n is a smooth function of n with an expansion in powers of n^{-p}. The literal sum $\frac{p}{n}\sum_{u_m<1}(1-u_m)^{-1/2}$, $u_m = 4cm^p/n^p$, differs from its Mellin pole part by an oscillation of order $n^{-1/2}$ set by the fractional part of $(n^p/4c)^{1/p}$, which is not present in b_n. Theorem 4.2 says that the band factor produces the pole part only. (d) The negative-axis geometry ($p > 1$) and the seam geometry ($p < 1$) give the same kernel. At $p = 1$ the two cotangents coincide and their poles merge, which is where the expansions of [10] change form. (e) The residue sum in (iii) is asymptotic only: $\zeta(-jp)$ grows factorially in j.

Numerically, the quadrature of e_n and the Mellin–Barnes integral agree to all 16 printed digits at $(p,n) = (1.5,10), (1.25,20), (3,10), (0.6,10), (0.3,8), (0.75,20)$, and the partial residue sums approach them (scripts `spec_kernel`, `m2_thmK_plt1`).

4.3. **Perturbed index functions.** For a node set other than the lattice, the analysis of [10] has not been carried out. The following computation shows what the band factor would contribute if the construction carried over with the lattice's index function $t(z) = nz^{-1/p}$ replaced by $t(z) = \kappa(n^p/z)$, κ the inverse of $k \mapsto \lambda_k$.

Proposition 4.4. *Let $p > 1$, $m > 0$, and for $\lambda_k = (k+\gamma k^{-m})^p$ let $\delta t_+(r) := -\gamma n^{-m}(re^{i\pi})^{m/p}$ be the first-order change of $t_+ = nr^{-1/p}e^{-i\pi/p}$ on the negative axis, and $\delta\mathfrak{e} := 2\,\mathrm{Im}[\mathcal{F}'(t_+)\delta t_+]$ with $\mathcal{F}'(t) = 2\pi i/(e^{2\pi i t}-1)$. Then, for $\mathrm{Re}\, w > (m+1)/p$,*

$$\int_0^\infty \delta\mathfrak{e}(-r) r^{-w-1}\,\mathrm{d}r = -2\pi\gamma p\, n^{-pw}\zeta(1+m-pw)\Big[\cos(\pi w) + \sin(\pi w)\tan\frac{\pi(pw-m)}{2}\Big].$$

The proof is the computation of Lemma 2.2: expand $\mathcal{F}'$ in powers of $e^{-2\pi i t_+}$, integrate termwise by [4, (5.9.1)], sum $\sum_{m'} m'^{m-pw} = \zeta(pw-m)$ by [4, (25.2.1)] and apply [4, (25.4.1)]; a quadrature check agrees to 10^{-13} at $w = 2.3$ and 3.1 ($p = 1.5$, $m = 1$). With $\delta Z(-w) = pw\gamma\zeta(1+m-pw)$, the first-order change of $Z_\Lambda(-w)$, the change of e_n is $\frac{1}{2\pi i}\int pW(w)\delta Z(-w) n^{-pw}\pi[-\cot(\pi w) - \tan(\pi(pw-m)/2)]\,\mathrm{d}w$ on a line $\mathrm{Re}\, w > (m+1)/p$.

Remark 4.5. Under that hypothetical construction, the integer residues give $pW(j)\delta Z(-j)n^{-jp}$, the zeta terms of the perturbed node set, exactly as for the lattice. The pole of $\delta Z(-w)$ at $q = m/p$ lies to the left of the line, so the band factor does not carry the pole term $pR_qW(q)\pi\cot(\pi q)n^{-1-pq}$ of Section 7; in the dictionary that term comes from the equilibrium, Proposition 3.5. The tan-pole at $w = (m+1)/p$ and the change $\log|t'|$ of the node density sit at order n^{-2-m}, the order of the subleading pole terms of Remark 7.2. None of this is a statement about b_n.

5. The natural index and the expansion of the lattice

This section proves Theorem A. The $1/n$ term and the $(s-1)^2$ part of the n^{-2} term of the lattice are one index shift (§5.1); for Bessel zeros the shift is exact to all orders (§5.2). We then extend the analysis of [10] and Theorem 2.5 to every order K in $1/n$ and every order B in the band-factor parameters (§§5.3–5.4). The analytic part is the order-K version of estimates already proved at order 2; the algebraic part is a finite residue recursion, which we evaluate in exact arithmetic. Strategy: the band factor is a tilt (Lemma 2.2), tilts act on the jump data by an entire gauge (Lemma 2.3), so every coefficient of the expansion is an entire function of the tilt parameters μ with n-independent Taylor coefficients, and the Taylor coefficients are computed by the recursion.

5.1. **The index shift.** Put $\delta_N := \frac{\alpha+1}{2}$, so that $N = n + \delta_N$, and

$$f_2 = f_2(p,\alpha) := \frac{p+1}{12} - \frac{(p+2)\alpha^2}{8}. \tag{5.1}$$

Proposition 5.1. *For all p, s, $d_1 = -p\delta_N$ and $d_2(p,s) = \frac{p(p+1)}{2}\delta_N^2 + f_2$. Equivalently, $1 + d_1/n + d_2/n^2 = (n/N)^p(1 + f_2N^{-2}) + O(n^{-3})$.*

Proof. With $s - 1 = p(\alpha+1)$ and $s - 1 - p = p\alpha$, the three terms of d_2 in Theorem 2.5 are $p(p+1)(\alpha+1)^2/8 = \frac{p(p+1)}{2}\delta_N^2$, $-(p+2)\alpha^2/8$ and $(p+1)/12$. Since $(1+\delta/n)^{-p} = 1 - p\delta/n + \frac{p(p+1)}{2}\delta^2 n^{-2} + O(n^{-3})$, the claim follows. □

So the lattice expansion to order n^{-2} is an index shift by δ_N followed by a bare function with no $1/N$ term. The shift has two parts: $\frac{\alpha}{2}$, attached to the hard edge of Bessel order α, and $\frac{1}{2} = -\zeta(0)$, the regularised count of the nodes (Section 7).

5.2. **Bessel zeros.** For $\mu > -1$ let $j_{\mu,1} < j_{\mu,2} < \cdots$ be the positive zeros of the Bessel function J_μ and $\lambda_k := (j_{\mu,k}/\pi)^2$. For $p = s = 2$ the measure (1.1) is $\sigma_\Lambda = \pi^2 \sum_k j_{\mu,k}^{-2} \delta_{\pi^2/j_{\mu,k}^2}$, and $\mu = \frac{1}{2}$ gives the lattice $\lambda_k = k^2$.

Proposition 5.2. *The recurrence coefficients of σ_Λ satisfy, for $n \geq 1$, with $N := n + \mu/2$,*

$$\frac{n^2 b_n}{c(2)} = \frac{4n^2}{(2n+\mu)\sqrt{(2n+\mu)^2 - 1}} = \Big(\frac{n}{N}\Big)^2 \Big(1 - \frac{1}{4N^2}\Big)^{-1/2}.$$

The bare function $(1 - \frac{1}{4N^2})^{-1/2} = 1 + \frac{1}{8N^2} + \frac{3}{128N^4} + \frac{5}{1024N^6} + \cdots$ is even in $1/N$, and its coefficients are $f_2(2, -\frac{1}{2})$, $f_4(2, -\frac{1}{2})$, $f_6(2, -\frac{1}{2})$ of (5.1), (5.3), (5.4).

Proof. The Lommel polynomials of order $\nu = \mu + 1$ satisfy the monic recurrence $x h_n = h_{n+1} + \beta_n^2 h_{n-1}$ with $\beta_n^2 = 1/(4(n+\nu-1)(n+\nu))$, and their Jacobi operator has spectrum the closure of $\{\pm 1/j_{\nu-1,k}\}$ [13, §3]; the orthogonality measure is symmetric with masses proportional to $j_{\nu-1,k}^{-2}$. By the quadratic transformation of symmetric measures (as in [10, Remark 9.6]), the image measure in $y = x^2$ has $b_n^{(y)} = \beta_{2n-1}\beta_{2n}$, and σ_Λ is its image under $y \mapsto \pi^2 y$. Hence $b_n = \pi^2/(4(2n+\mu)\sqrt{(2n+\mu)^2-1})$, and $c(2) = \pi^2/16$ gives the first identity. The second is $(2N)\sqrt{4N^2-1} = 4N^2(1 - \frac{1}{4N^2})^{1/2}$. □

The first identity agrees with the Jacobi coefficients computed from the zeros (mpmath zeros for $k \leq 120$, McMahon's expansion beyond, a continuous tail) to $1.6 \cdot 10^{-15}$ at $\mu = 0$ and $\mu = 1.5$, $n \leq 40$ (script `spec_lommel`). Here $\alpha = (s-1-p)/p = -\frac{1}{2}$, and the large-$k$ form $j_{\mu,k} = \pi(k + \frac{\mu}{2} - \frac{1}{4}) + O(1/k)$ gives $Z_\Lambda(0) = \zeta(0, \frac{\mu}{2} + \frac{3}{4}) = -\frac{\mu}{2} - \frac{1}{4}$, so $N = n + \frac{\alpha}{2} - Z_\Lambda(0)$: Proposition 5.2 is the natural-index law of Section 7, exactly.

5.3. **The residue recursion.** We use the frame of §2.2 for $p \neq 1$: the clockwise circles ∂U_0, ∂U_r around 0 and $4c$, the local maps f (hard edge) and $\tilde{\varphi}$ (soft edge), and the Szegő quotients ρ, ω of [10, Sections 8–10]. Let $M := \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & i \\ i & 1 \end{smallmatrix}\right)$, let $\circ$ be the entrywise product, let $a_k = a_k(\alpha)$, $b_k = b_k(\alpha)$ be the coefficients of the Bessel expansions [4, (10.17.1), (10.17.8)] and u_k, v_k those of the Airy expansions [4, (9.7.2)], and put

$$\Xi_k := M^{-1}(M \circ C_k),\ C_k := \begin{pmatrix} (-1)^k a_k & a_k \\ (-1)^k b_k & b_k \end{pmatrix}, \qquad X_k := M^{-1}(M \circ C_k'),\ C_k' := \begin{pmatrix} u_k & (-1)^k u_k \\ v_k & (-1)^k v_k \end{pmatrix}.$$

Lemma 5.3 (the jumps to every order)**.** *Let $p \neq 1$ and $K, J \geq 1$. Uniformly on $\partial U_0 \cup \partial U_r$, boundary values included, the jump of the error matrix on the circles is $I + \sum_{k \leq K} \Delta_k n^{-k} + O(n^{-K-1})$, where Δ_k equals*

$$\Delta_k^0 = (2f^{1/2})^{-k} P_0^\infty \rho^{\sigma_3} (\sigma_3 \Xi_k \sigma_3) \rho^{-\sigma_3} (P_0^\infty)^{-1} \qquad \text{on } \partial U_0,$$
$$\Delta_k^r = (-1/\tilde{\varphi})^k P_0^\infty (-\omega)^{-\sigma_3/2} X_k (-\omega)^{\sigma_3/2} (P_0^\infty)^{-1} \qquad \text{on } \partial U_r,$$

and $\Delta_k = G_{\mu_n} \Delta_k^{\mathrm{mod}} G_{\mu_n}^{-1} + O(n^{-(J+1)p})$.

Proof sketch. For $k = 1$ this is [10, Lemmas 9.1, 9.2 and 10.6], and for $k = 2$ the third-order version cited in §2.4. The proof is the same with $K+1$ terms of the expansions [4, (10.40.1)–(10.40.4), (10.17.5), (10.17.6), (10.17.11), (10.17.12)] of the Bessel model [7, §6] and [4, (9.7.5)–(9.7.6)] of the Airy model, whose Poincaré remainders are $O(n^{-K-1})$ uniformly because the arguments are of size n on the circles. The sign pattern $(-1)^k$ is the same in every sector, for every k: each column of the Airy model is a constant times $\mathrm{Ai}(Z)$ or $\mathrm{Ai}'(Z)$ with $Z = e^{2\pi i m/3}\zeta$, $|\mathrm{ph}\, Z| \leq \pi - \delta$, and $Z^{3/2} = \pm\zeta^{3/2}$, so $(-1)^k$ sits on the column that carries $e^{-\xi}$ whichever branch occurs; the column carrying $e^{\pm\xi}$ is fixed at $k = 0$. The Bessel model

is the same with $Y = \pm w$. The n-dependence of the weight is Lemma 2.2 at order J, and the conjugation by G_{μ_n} is Lemma 2.3, whose proof commutes diagonal factors and does not depend on k. □

Lemma 5.4 (meromorphy)**.** *For every k, Δ_k^{mod} is meromorphic in U_0 and in U_r, and $\Delta_k^{\mu} := G_\mu \Delta_k^{\mathrm{mod}} G_\mu^{-1}$ is meromorphic with Laurent coefficients that are entire functions of μ.*

Proof sketch. For $p > 1$: $\hat{P}^{\mathrm{mod}}(P_0^\infty)^{-1}$ is analytic in $U_e \setminus \{e\}$ and has the expansion of Lemma 5.3 uniformly on compact subsets, so $\Delta_k = \lim n^k(\cdots)$ is analytic and single-valued there; its explicit formula grows at most polynomially at e, so e is a pole. For $0 < p < 1$, where the glued hard-edge parametrix carries seam jumps inside U_0, one checks directly that Δ_k has no jump: from the definitions, $X_k = \frac{1}{2}(u_k + v_k)I - \frac{1}{2}(u_k - v_k)\sigma_2$ for even k and $X_k = \frac{1}{2}(u_k + v_k)\sigma_3 + \frac{1}{2}(v_k - u_k)i\sigma_1$ for odd k, so $JX_kJ^{-1} = (-1)^k X_k$ for $J = i\sigma_2$; on $(4c - \epsilon, 4c)$, $\tilde{\varphi}_+ = -\tilde{\varphi}_-$ contributes the same $(-1)^k$, and $P_{0,+}^\infty = P_{0,-}^\infty J$, $(-\omega)_+^{1/2}(-\omega)_-^{1/2} = 1$. The hard edge is the same with $\sigma_3 \Xi_k \sigma_3$, $f_+^{1/2} = -f_-^{1/2}$ and the relations for ρ of [10, Section 10]. The gauge G_μ is entire in (z, μ) and satisfies the same relations. □

For $\mu \in \mathbb{C}^J$ define, near each $e \in \{0, 4c\}$, $\mathsf{R}_{\mathrm{in}}^{(0)} := I$ and for $m \geq 1$

$$\begin{aligned} \mathsf{F}_m(\mu) &:= \sum_{j=1}^{m} \mathsf{R}_{\mathrm{in}}^{(m-j)}(\mu)\Delta_j^{\mu}, \qquad \mathsf{R}_{\mathrm{out}}^{(m)}(\mu) := \sum_{e \in \{0,4c\}} \big(\text{principal part of } \mathsf{F}_m(\mu) \text{ at } e\big), \\ \mathsf{R}_{\mathrm{in}}^{(m)}(\mu) &:= \mathsf{R}_{\mathrm{out}}^{(m)}(\mu) - \mathsf{F}_m(\mu), \qquad \mathsf{R}_1^{(m)}(\mu) := \mathrm{Res}_0\, \mathsf{F}_m(\mu) + \mathrm{Res}_{4c}\, \mathsf{F}_m(\mu). \end{aligned} \tag{5.2}$$

By Lemma 5.4 these are well defined, $\mathsf{R}_{\mathrm{in}}^{(m)}$ is analytic in each disc, and all entries are entire in μ. For $m = 1, 2$ and $\mu = 0$ they are the objects of [10, Proposition 9.5] and its third-order version.

Proposition 5.5 (small norm to order K)**.** *Let $p \neq 1$ and $K, J \geq 1$. Then $\mathsf{R}_1 = \sum_{m \leq K} \mathsf{R}_1^{(m)}(\mu_n) n^{-m} + O(n^{-K-1} + n^{-1-(J+1)p})$, and the same expansion holds for R uniformly at positive distance from its jump contour.*

Proof sketch. Write $\eta_n := n^{-K-1} + n^{-1-(J+1)p}$. By Lemma 5.3 the jump is $I + \sum_{k \leq K} \Delta_k^{\mu_n} n^{-k}$ on the circles up to $O(\eta_n)$, and off the circles it is I up to errors that are $O(\eta_n)$ in $L^1 \cap L^2 \cap L^\infty$, as in [10, Propositions 9.5 and 10.8]; the Cauchy operator is bounded uniformly in n. One proves by induction on $m = 1, \ldots, K+1$ that the boundary value R_- equals $\sum_{k<m} \mathsf{R}_-^{(k)} n^{-k} + O_{L^2}(n^{-m} + n^{-1-(J+1)p})$, where $\mathsf{R}^{(k)}$ is $\mathsf{R}_{\mathrm{in}}^{(k)}(\mu_n)$ inside the discs and $\mathsf{R}_{\mathrm{out}}^{(k)}(\mu_n)$ outside. In the step, insert the claim into $\mathsf{R} = I + \mathcal{C}(\mathsf{R}_-(J - I))$: the new error is $O(n^{-m-1} + n^{-1-(J+1)p})$, and the term of order n^{-m} is the Cauchy integral over the clockwise circles of $\mathsf{F}_m(\mu_n)$, which by the residue theorem is $\mathsf{R}_{\mathrm{out}}^{(m)}$ outside the discs and $\mathsf{R}_{\mathrm{out}}^{(m)} - \mathsf{F}_m = \mathsf{R}_{\mathrm{in}}^{(m)}$ inside. The coefficient R_1 is $-\frac{1}{2\pi i} \int \mathsf{R}_-(J - I)$, and $-\frac{1}{2\pi i} \oint_{\mathrm{cw}} \mathsf{F}_m = \mathrm{Res}_0\, \mathsf{F}_m + \mathrm{Res}_{4c}\, \mathsf{F}_m$. □

Define $\mathfrak{D}_K(n; \mu) := ((Q_1)_{12}(Q_1)_{21})^{1/2}/c$, expanded to order n^{-K}, where $Q_1 = P_1 + \sum_{m \leq K} \mathsf{R}_1^{(m)}(\mu) n^{-m}$, and write $\mathfrak{D}_K(n; \mu) = \sum_\beta \mu^\beta \mathfrak{D}_{\beta,K}(n)$ for its Taylor series in μ. Each $\mathfrak{D}_{\beta,K}$ is a polynomial in $1/n$ of degree at most K. A polynomial in $1/n$ is *written in natural form* when it is re-expanded in $1/N$ by $1/n = N^{-1}/(1 - \delta_N N^{-1})$; a layer μ^β carries $n^{-|\beta|_w p} = N^{-|\beta|_w p}(N/n)^{|\beta|_w p}$, with $|\beta|_w := \sum_j j\beta_j$.

Lemma 5.6 (exact algebra)**.** *The recursion (5.2) and the extraction $\mathfrak{D}_K$ give, identically in the symbols:*

(a) $\mu = 0$, $K = 6$: $\mathfrak{D}_{0,6}(n) = (n/N)^p[1 + f_2N^{-2} + f_4N^{-4} + f_6N^{-6} + O(N^{-7})]$, *with no* N^{-1}, N^{-3}, N^{-5} *terms,* f_2 *as in* (5.1) *and* f_4, f_6 *as in* (5.3), (5.4)*; in particular* $d_1 = -p(\alpha+1)/2$ *and* d_2 *is that of Theorem 2.5;*

(b) *first order in* μ_j*, for every integer* $j \geq 1$ *(*j *a symbol),* $K = 4$*: the coefficient of* n^{-1} *per unit* $\binom{2j}{j}\mu_j$ *is* $pj/2$ *(with* $c = 1$*), and in natural form the layer is a multiple of* $N^{-1-jp}[1+g_jN^{-2}+O(N^{-4})]$*, with no* N^{-1}*,* N^{-3} *term in the bracket and* $g_j = A_j+B_j\alpha^2$ *as in Theorem A; the same holds at* $K = 5$ *for the integers* $j = 1, \dots, 5$*, and for* $j = 1$ *the* N^{-4} *coefficient of the bracket is* $g_{1,4}$ *of* (5.5)*;*

(c) *second order in* μ_1*,* $K = 5$*: per unit* μ_1^2 *the natural form has no* N^{-1-2p} *term, the coefficient* $p(3p+1)/2$ *at* N^{-2-2p}*, no* N^{-3-2p} *and no* N^{-5-2p} *term, and* p^2h_2 *at* N^{-4-2p}*, with* h_2 *as in* (5.6)*.*

Proof. With $c = 1$ every input is a rational function of (p, α): the local maps have convergent series with rational coefficients, π cancels in f, the square roots are of series whose leading coefficient is 1 or a perfect square, and the factors $2^{-1/2}$ in T and M occur in conjugate pairs. The recursion is carried out on truncated Laurent series over $\mathbb{Q}(p, \alpha)[i]$; every series carries the index of its highest exactly known coefficient, and every principal part checks that its coefficients down to the residue are known, otherwise the run stops. For (b), μ_j is normalised by $\binom{2j}{j}$; the gauge to first order is $\partial_{\mu_j}\Delta_k^{\mu}|_{\mu=0} = [\mathcal{G}_j, \Delta_k^{\mathrm{mod}}]$ with $\mathcal{G}_j = \mathsf{T}\left(\begin{smallmatrix} 0 & -(z-4c)Q_j/2 \\ -zQ_j/2 & 0 \end{smallmatrix}\right)\mathsf{T}^{-1}$, and it needs only the Taylor data of $Q_j/\binom{2j}{j}$ at 0 and at $4c = 4$, which are rational in j: $\prod_{i\leq m}(j-i)/(2(2j-2i-1))$ at 0 and $j(j-1)\cdots(j-m)/(4^{m+1}(\frac{1}{2})_{m+1})$ at 4, from $\sum_j Q_j(z)t^j = t/((1-zt)\sqrt{1-4t})$. So (b) is a computation over $\mathbb{Q}(p, \alpha, j)[i]$. For (c) the gauge is taken to second order, $\cosh h = 1 + z(z-4)\mu^2Q_1^2/8 + O(\mu^4)$, $S = \mu Q_1/2 + O(\mu^3)$. The scripts and their outputs are listed in Appendix A; each output begins with the SHA-256 hashes of the scripts that produced it. Since $(z - 4c)Q_j$ and zQ_j scale as c^j under $z = c\zeta$, μ_j enters only as $c^j\mu_j$; the integer layer and the ratios g_j, h_2 are independent of c. □

The closed forms are

$$f_4 = \frac{(p+1)(6p^3+46p^2+55p+3)}{1440\,p(3p+2)} - \frac{5(p+1)(p+2)\alpha^2}{96(3p+2)} - \frac{p(p+2)^2\alpha^4}{128(3p+2)}, \tag{5.3}$$

$$f_6 = \frac{\phi_0 + \phi_2\alpha^2 + \phi_4\alpha^4 + \phi_6\alpha^6}{(3p+2)^2(5p+2)}, \tag{5.4}$$

$$\begin{aligned}
\phi_0 &= -\frac{p+1}{362880\,p^2}\big(450p^8 + 1590p^7 - 316p^6 - 23761p^5 - 66864p^4 - 65507p^3 \\
&\qquad - 18220p^2 - 1632p - 40\big), \\
\phi_2 &= \frac{(p+1)(p+2)(90p^6+270p^5+664p^4-165p^3-1649p^2-288p+28)}{11520\,p}, \\
\phi_4 &= -\frac{(p+1)(p+2)^2(20p^4+45p^3+162p^2+72p+16)}{1536}, \\
\phi_6 &= \frac{p(p+2)^3(16p^3+7p^2-4p-4)}{3072},
\end{aligned}$$

$$g_{1,4} = \frac{p+1}{5760\,p^2(3p+2)}\big(9p^6+117p^5+161p^4-569p^3-1550p^2-976p+48\big) + \frac{(p+1)(p+2)(3p^4+33p^3+124p^2+182p+108)\alpha^2}{192\,p(3p+2)} + \frac{(p+2)^3(p^3-4p-4)\alpha^4}{128\,p(3p+2)}, \tag{5.5}$$

$$h_2 = \frac{(p+1)(p+2)(3p^2+2p+1)}{8p^2} + \frac{(p+2)(2p^3-p^2-7p-6)\alpha^2}{16p^2}. \tag{5.6}$$

For $j=1$, $g_1 = (p+1)(p^2+3p-2)/(24p) + (p+2)(p^2+2p+2)\alpha^2/(8p)$. At the Bessel point $(p,\alpha)=(2,-\frac12)$, $f_4=\frac{3}{128}$ and $f_6=\frac{5}{1024}$, in agreement with Proposition 5.2.

5.4. **The theorem.**

Theorem 5.7. *Let $p>0$, $p\neq 1$, $s>1$, and $K,J,B\geq 1$.*

(i) Double expansion. $\dfrac{n^p b_n}{c} = \displaystyle\sum_{|\beta|\leq B} \mu_n^\beta \mathfrak{D}_{\beta,K}(n) + O\big(n^{-K-1} + n^{-2-(B+1)p} + n^{-1-(J+1)p}\big)$.

(ii) Natural forms of the layers. *$\mathfrak{D}_{0,6}$, $\mu_{n,j}\mathfrak{D}_{e_j,4}$ and $\mu_{n,1}^2\mathfrak{D}_{2e_1,5}$ satisfy*

$$\mathfrak{D}_{0,6}(n) = \Big(\frac{n}{N}\Big)^p\Big[1+\frac{f_2}{N^2}+\frac{f_4}{N^4}+\frac{f_6}{N^6}+O(N^{-7})\Big],$$
$$\mu_{n,j}\mathfrak{D}_{e_j,4}(n) = \Big(\frac{n}{N}\Big)^p\frac{\varepsilon_j}{N^{1+jp}}\Big[1+\frac{g_j}{N^2}+O(N^{-4})\Big] \qquad (j\geq 1),$$
$$\mu_{n,1}^2\mathfrak{D}_{2e_1,5}(n) = \Big(\frac{n}{N}\Big)^p\frac{\varepsilon_1^2}{N^{2+2p}}\Big[\frac{3p+1}{2p}+\frac{h_2}{N^2}+O(N^{-4})\Big],$$

with the brackets free of odd powers of $1/N$ to the orders shown; for $j=1$ the second bracket is $1+g_1N^{-2}+g_{1,4}N^{-4}+O(N^{-5})$.

(iii) The coefficients of b_n. *With $K=6$, $B=2$ and $J\geq 6/p-1$, expansion (1.9) holds.*

Proof. (i) By (2.1) and Proposition 5.5, $n^{2p}b_n^2 = (Q_1)_{12}(Q_1)_{21}$ with $Q_1 = P_1 + \sum_{m\leq K}\mathsf{R}_1^{(m)}(\mu_n)n^{-m} + O(n^{-K-1}+n^{-1-(J+1)p})$. The n^0 term of $(Q_1)_{12}(Q_1)_{21}$ is c^2, since P_1 does not depend on μ, so each coefficient of $\mathfrak{D}_K(n;\mu)$ is a polynomial in the entries of the $\mathsf{R}_1^{(i)}(\mu)$, $i\leq K$, which are entire in μ and independent of n. Taylor's theorem in μ on $|\mu|\leq 1$ applies coefficientwise. The n^0 coefficient is 1 for all μ, and the n^{-1} coefficient is $d_1+cpM_\mu(4c)$, exactly linear in μ by Lemma 2.4; hence $\mathfrak{D}_{\beta,K}=O(n^{-2})$ for $|\beta|\geq 2$, and the Taylor remainder after order B is $O(|\mu|^{B+1}n^{-2})$. Since $|\mu_n|=O(n^{-p})$, this is $O(n^{-2-(B+1)p})$.

(ii) This is Lemma 5.6 in natural form, with the normalisations $\mu_{n,j}\binom{2j}{j}pc^j j/2 = \varepsilon_j n^{-jp}$ and $\mu_{n,1}^2 p(3p+1)/2\cdot c^2 = \varepsilon_1^2 n^{-2p}(3p+1)/(2p)$.

(iii) In (i) keep $\beta=0$ at $K=6$; $\beta=e_j$ for every j with $1+jp<\min(7,2+3p)$, at $K=5$ ($j\leq 5$) or $K=4$; and $\beta=2e_1$ at $K=5$. The other terms with $|\beta|=2$ ($\mu_1\mu_2,\mu_2^2,\dots$) are $O(n^{-2-3p})$ because $\mathfrak{D}_\beta=O(n^{-2})$ and $|\beta|_w\geq 3$, and those with $|\beta|=3$ are $O(n^{-2-3p})$. The truncation errors are bounded as follows. The $j=1$ bracket is known to relative order N^{-5}, an error $O(n^{-6-p})$, which is $O(n^{-2-3p})$ for $p<2$ and $O(n^{-8})$ for $p\geq 2$. For $j\geq 2$ the bracket error is $O(n^{-5-jp})$, and $5+2p>2+3p$ for $p<3$, $5+2p\geq 11$ for $p\geq 3$. The h_2 term of the ε_1^2 layer is $O(n^{-4-2p})$, which is $O(n^{-2-3p})$ for $p<2$ and $O(n^{-7})$ for $p\geq\frac32$. Finally $n^{-1-(J+1)p}=O(n^{-7})$. □

A term CN^{-e} of (1.9) is a coefficient of the expansion of $n^p b_n/c$ (in powers of N, or equivalently, after re-expansion, of n) exactly when $e < \min(7, 2+3p)$. Table 5.1 lists the resulting ranges. The terms outside their range are true identities of the layers in Theorem 5.7(ii) but are not determined as coefficients of b_n, because the uncomputed layers $\mu_1\mu_2$, μ_1^3 enter at n^{-2-3p}.

term	exponent e	a coefficient of $n^p b_n/c$ for
$f_2 N^{-2}$	2	all $p \neq 1$
$f_4 N^{-4}$	4	$p > \frac{2}{3}$
$f_6 N^{-6}$	6	$p > \frac{4}{3}$
$\varepsilon_j N^{-1-jp}$	$1+jp$	$(j-3)p < 1$ and $1+jp < 7$
$\varepsilon_1 g_1 N^{-3-p}$	$3+p$	$\frac{1}{2} < p < 4$
$\varepsilon_2 g_2 N^{-3-2p}$	$3+2p$	$1 < p < 2$
$\varepsilon_1 g_{1,4} N^{-5-p}$	$5+p$	$\frac{3}{2} < p < 2$
$\frac{3p+1}{2p}\varepsilon_1^2 N^{-2-2p}$	$2+2p$	$p < \frac{5}{2}$
$\varepsilon_j g_j N^{-3-jp}$ $(j \geq 3)$, $\varepsilon_1^2 h_2 N^{-4-2p}$		never at $B = 2$

TABLE 5.1. Which terms of (1.9) are coefficients of b_n ($p \neq 1$). Coincident exponents merge: at $p = 1/j_0$, $\varepsilon_{j_0} N^{-2}$ merges with $f_2 N^{-2}$, and $f_2 + \varepsilon_{j_0} = f_2 - p/12$, as in Theorem 2.5.

Remark 5.8. (a) Theorem 5.7 does not prove evenness in $1/N$ to all orders, nor any closed form of the bare function; the computed layers are listed in (ii), and the layers $\mu_1\mu_2$, μ_2^2, μ_1^3 and higher are not computed. Nothing is claimed at $p = 1$, where the hard-edge and band-factor constructions of [10] change. (b) The same recursion evaluated numerically at 40 digits, with truncation depths 40–64, gives $f_1 = f_3 = f_5 = f_7 = 0$ to 10^{-35} at every sampled (p, α) with $p \in [\frac{1}{3}, 12]$, each f_{2k} even in α, and $f_8(3,0) = 1311857/632311515$, $f_8(2,0) = 3899/2^{21}$ after rational recognition. The denominators of f_4, f_6 are $3p+2$ and $(3p+2)^2(5p+2)$; they come from the factors $1/(a'+k)$, $a' = \frac{1}{p} + \frac{1}{2}$, of the hard-edge local map, $a'+k = ((2k+1)p+2)/(2p)$, and $632311515 = 3^5 \cdot 5 \cdot 11^3 \cdot 17 \cdot 23$ is $3^5 \cdot 5\,(3p+2)^3(5p+2)(7p+2)$ at $p = 3$ (scripts `m4_rh4`, `m4_f8_denominator`). These statements are numerical. (c) Heuristically, evenness in α after the shift reflects the fact that the Bessel coefficients $a_k(\alpha)$, $b_k(\alpha)$ depend on α only through α^2, while elsewhere α enters only through the power of z at the hard edge, which the shift by $\frac{\alpha}{2}$ absorbs. We have no proof, and no explanation of the evenness in $1/N$. (d) The factor $1/(2j-1)$ in g_j, read as a function of a continuous Mellin variable w, is a simple pole at $w = \frac{1}{2}$, the Mellin signature of $(1-u)^{+1/2}$, against $(1-u)^{-1/2}$ for the leading kernel (Lemma 4.1). Partial fractions give $B_j = \frac{p+2}{8p}\left[\frac{p^2 j}{2} + \frac{p(p+2)}{4} + \frac{(p+2)(p+4)}{4(2j-1)}\right]$. This suggests a second-order edge kernel (Problem 7.8).

6. Examples and numerical evidence

This section tests Theorem 5.7 on the lattice and collects the evidence for the conjectures of Section 7 on other node sets. No proof uses it. The computations are floating-point; the scripts and their stored outputs are listed in Appendix A.

Method. The coefficients b_n, $n \leq 1600$, are computed in quadruple precision by adding the atoms $1/\lambda_m$, $m \leq K$, one at a time and restoring tridiagonal form by plane rotations, with a Gauss–Jacobi rule for the continuous tail beyond K; K is 100 to 200 times the largest n. Expansion coefficients are obtained by least-squares fits of n-windows ($n \geq 200$, 400, 800 or mid-range windows) after subtracting the terms that are known exactly, with a pool of nuisance

exponents; the number in parentheses is the spread of the fitted value over windows and pools. When a prediction was formed after looking at some of the data, we say which rows were used to form it; those rows are not evidence for it.

6.1. **The lattice in natural form.** Table 6.1 maps the lattice coefficients to $F(N) := (N/n)^p n^p b_n/c$, subtracts $1 + f_2 N^{-2} + \sum_j \varepsilon_j N^{-1-jp}$ and fits the remainder. The N^{-1} and N^{-2} residuals re-confirm d_1 and d_2. The N^{-3} residual is zero to the precision of the tail windows, against lattice n^{-3} coefficients of size 0.1 to 1 (for example -0.267 at $p = 1.5$, $s = 2$); its sign follows α at $p = 1.5$, but the large-n end of the data is noise-dominated at this scale, and mid-range windows $n \in [10, 400]$ give N^{-3} coefficients below $4 \cdot 10^{-8}$ in all cases of Table 6.2. Theorem 5.7 proves the absence of the N^{-3} term for $p > 1$.

p	s	α	N^{-1} residual	N^{-2} residual	N^{-3} residual
1.5	2	$-\frac{1}{3}$	$-4 \cdot 10^{-11}$	$+1.5 \cdot 10^{-7}$	$-3.0 \cdot 10^{-4}$ $(2 \cdot 10^{-4})$
1.5	3	$+\frac{1}{3}$	$+5 \cdot 10^{-11}$	$-1.6 \cdot 10^{-7}$	$+3.2 \cdot 10^{-4}$ $(2 \cdot 10^{-4})$
1.25	2	-0.2	$+4 \cdot 10^{-11}$	$-1.5 \cdot 10^{-7}$	$+5.2 \cdot 10^{-4}$ $(4 \cdot 10^{-4})$
2.5	2	-0.6	$-2 \cdot 10^{-11}$	$+5 \cdot 10^{-8}$	$-6 \cdot 10^{-5}$ $(6 \cdot 10^{-5})$
3	2	$-\frac{2}{3}$	$+2 \cdot 10^{-12}$	$-4 \cdot 10^{-9}$	$+2.3 \cdot 10^{-6}$ $(5 \cdot 10^{-7})$
1.5	2.5	0	$+2 \cdot 10^{-12}$	$-7 \cdot 10^{-9}$	$+1.3 \cdot 10^{-5}$ $(2 \cdot 10^{-5})$
2	3	0	$-2 \cdot 10^{-13}$	$+4 \cdot 10^{-10}$	$-3 \cdot 10^{-7}$ $(3 \cdot 10^{-7})$

TABLE 6.1. Residual coefficients of $F(N) - 1 - f_2 N^{-2} - \sum_j \varepsilon_j N^{-1-jp}$ for the lattice, tail windows $n \geq 200, 400$ (script `m4_natural_index`).

p	α	f_4 fitted	f_4, (5.3)	difference
2	0	0.044921952 $(1 \cdot 10^{-6})$	0.044921875	$7.7 \cdot 10^{-8}$
2	1	-0.064452710 $(2 \cdot 10^{-6})$	-0.064453125	$4.2 \cdot 10^{-7}$
4	0	0.083271330 $(3 \cdot 10^{-7})$	0.083271329	$2.9 \cdot 10^{-10}$
4	1	-0.108692879 $(9 \cdot 10^{-7})$	-0.108692956	$7.7 \cdot 10^{-8}$
2.5	0.5	0.029254295 $(1 \cdot 10^{-4})$	0.029206272	$4.8 \cdot 10^{-5}$

TABLE 6.2. The N^{-4} coefficient of the lattice, mid-range windows (script `m4_f4_lattice`). The polynomial $f_4(2, \alpha) = (23 - 40\alpha^2 - 16\alpha^4)/512$ was first read off fits of this kind; the exact computation of Lemma 5.6 was built independently.

For the zeta layer, after subtracting the integer layer and the bare terms $\varepsilon_j N^{-1-jp}$, mid-range fits at $p \in \{1.5, 2.5\}$ give an N^{-2-p} coefficient below $4 \cdot 10^{-7}$ and an N^{-3-p} coefficient equal to $\varepsilon_1 g_1$ within $4 \cdot 10^{-6}$ to $6 \cdot 10^{-5}$, with spreads $8 \cdot 10^{-4}$ to 10^{-2} (script `m4_zeta_lattice`); at $p = 1.5$ the values $g_1 = 95/288$ ($\alpha = 0$) and $22/9$ ($\alpha = 1$) were first seen in these fits.

6.2. **Hurwitz lattices.** For $\lambda_m = (m+\delta)^p$, $\delta > -1$, one has $Z_\Lambda(w) = \zeta(pw, a)$ with $a = 1+\delta$, so $Z_\Lambda(-j) = \zeta(-jp, a)$ and $Z_\Lambda(0) = \frac{1}{2} - a$. The natural-index law of Section 7 predicts the $1/n$ coefficient $d_1 = -\frac{s-1}{2} - p\delta$ and the n^{-1-p} coefficient $2pc\,\zeta(-p, a)$ in place of k_p. Table 6.3 compares both with fits; the lattice value k_p is excluded in every row.

The n^{-2} coefficients of Hurwitz lattices at $p = 0.6$ ($a = \frac{1}{2}, \frac{3}{2}, 2$), 0.75 and 1.5 agree with the index-shifted lattice value $d_2 + \frac{p(p+1)}{2}\delta^2 - (p+1)\delta d_1$ to between $2 \cdot 10^{-6}$ and $3 \cdot 10^{-5}$ (spreads

p	s	a	d_1 fitted	d_1 predicted	n^{-1-p} fitted	$2pc\,\zeta(-p,a)$
0.6	2	$\frac{1}{2}$	-0.19999996	-0.2	$+0.0306156$ $(8\cdot10^{-5})$	$+0.0306142$
0.6	2	$\frac{3}{2}$	-0.80000024	-0.8	-0.309374 $(1.4\cdot10^{-4})$	-0.309386
0.6	2	2	-1.10000058	-1.1	-0.605309 $(3\cdot10^{-4})$	-0.605320
0.75	3	2	-1.75000033	-1.75	-0.779282 $(1.3\cdot10^{-3})$	-0.779293
1.5	2	$\frac{3}{2}$	-1.25000000	-1.25	-0.57244 $(5.5\cdot10^{-3})$	-0.57263
1.5	2	2	-2.00000000	-2	-1.74157 $(5.6\cdot10^{-3})$	-1.74211

TABLE 6.3. Hurwitz lattices (script `spec_hur`). The lattice values of the n^{-1-p} coefficient are -0.0899767 ($p=0.6$), -0.0918692 ($p=0.75$) and -0.0432947 ($p=1.5$).

$3\cdot10^{-5}$ to $9\cdot10^{-4}$), for shifts of size 0.3 to 3 (script `spec_hur_d2`). At $p=1.5$ the rule "the lattice expansion as a function of $n+\delta$, multiplied by $(n/(n+\delta))^p$, with every $\zeta(-jp)$ replaced by $\zeta(-jp,a)$" also predicts the n^{-3} coefficients ($\delta=-\frac12,\frac12,1$; agreement within the fit ranges, $1\cdot10^{-4}$ to $1.3\cdot10^{-2}$, relative $3\cdot10^{-4}$ to $2\cdot10^{-3}$ of the shifts; this uses the fitted lattice d_3) and the $n^{-3.5}$ coefficients, where the alternative "zeta terms unshifted" is excluded at $\delta=\frac12,1$ (script `spec_shift_hi`). At $p=\frac13$ and $\frac14$ ($\delta=-\frac12$) the first two, resp. three, coefficients $2pc\,\zeta(-p,a)$, $6pc^2\zeta(-2p,a)$, $20pc^3\zeta(-3p,a)$ agree to between $4\cdot10^{-7}$ and $3\cdot10^{-4}$, and at $p=\frac13$ the merged n^{-2} coefficient requires the term $\frac13\zeta(-1,a)$: without it the fit is off by seven spreads (script `spec_hur_small`).

6.3. **The pole law on synthetic lattices.** For $\lambda_k=(k+\gamma k^{-m})^p$ the counting function is unchanged at order k^0, and to first order in γ, $Z_\Lambda(-w)=\zeta(-pw)+pw\gamma\zeta(m+1-pw)+O(\gamma^2)$ has a new pole at $q=m/p$ with residue $R=-m\gamma/p$. The pole law (Conjecture 7.1(c)) predicts the term $pRW(q)\pi\cot(\pi q)n^{-1-pq}=-m\gamma W(m/p)\pi\cot(\pi m/p)n^{-1-m}$ in n^pb_n/c. Table 6.4 compares it with fits of $n(y_n-y_n^{\mathrm{lat}})$ after subtraction of the first-order zeta terms $pW(j)\delta Z(-j)n^{-jp}$. For $m=1$, $W(1/p)=1$, so the rows test the cotangent; the rows $m=\frac12$ test W as well.

p	m	$\cot(\pi m/p)$	predicted	fitted, $n\ge200$ / 400 / 800	role
0.6, 1.5	1	-0.5774	$+0.906900$	$+0.906900$	identification
1.25	1	-1.3764	$+2.162016$	$+2.162016$ / $+2.162017$ / $+2.162022$	prediction
1.8	1	-0.1763	$+0.276974$	$+0.276974$ / $+0.276974$ / $+0.276973$	prediction
0.8	1	$+1$	-1.570796	-1.570783 / -1.570792 / -1.570813	prediction
0.75	1	$+0.5774$	-0.906900	-0.906866 / -0.906885 / -0.906954	prediction
1.5	$\frac12$	$+0.5774$	-0.424704	-0.424703 / -0.424697 / -0.424630	prediction
1.25	$\frac12$	$+0.3249$	-0.235183	-0.235177 / -0.235112 / -0.232695	prediction

TABLE 6.4. The pole law, $s=2$, $\gamma=\frac12$ (scripts `spec_pole`, `spec_gamma`). The kernel $\pi\cot(\pi w)$ was identified from the first row; the predictions of the other rows were fixed before the corresponding measures were computed. For $m=\frac12$, $W(m/p)=0.9366$ ($p=1.5$) and 0.9216 ($p=1.25$). The $n\ge800$ windows drift where few points remain.

A constant shift. For $\lambda_k=k^p+\kappa$ the index offset grows like k^{1-p}, and the expansion changes at relative order n^{-p}, not n^{-1-p}. To first order, $Z_\Lambda(-w)=\zeta(-pw)+\kappa w\zeta(p(1-w))$ has its Weyl pole moved to $q=1-1/p$, with residue $-\kappa q/p$, and the pole law gives

$-2(2-p)c\kappa\, n^{-p}$ in $n^p b_n/c$, using $W(-1/p) = p\tan(\pi/p)/\pi$. Fits of the n^{-p} coefficient give -1.20342 (0.13), -0.60154 (0.04), -1.22311 (0.74) at $(p,\kappa) = (0.6,1), (0.6,\frac{1}{2}), (0.3,1)$, against -1.20247, -0.60123, -1.22262 (script `spec_kappa_fit`); the first row was the identification point, and at $p = 0.3$ the spread is 60% of the value, so that row is not evidence. The Hankel determinants of the shifted measure are those of the old nodes with an m-dependent weight: since $\varphi(y) = y/(1+\kappa y)$ satisfies $\varphi(a) - \varphi(b) = (a-b)/((1+\kappa a)(1+\kappa b))$, Heine's formula gives $D_m(\sigma_\kappa) = D_m((1+\kappa y)^{-2(m-1)}\sigma')$ with $\sigma' = (1+\kappa y)^{-s/p}\sigma$, and a first-order expansion of $\log D_m$ in κ reproduces the law, with a relative gap that decays like κn^{-p} (script `spec_mobius_pred`).

6.4. **The subleading pole term.** The pole law gives the leading power n^{-1-pq} of a pole. The same pole contributes at n^{-2-pq}, and this term is not removed when $\cot(\pi q) = 0$ (Remark 7.2). For the family of §6.3, the part linear in γ of the n^{-2-m} coefficient of $n^p b_n/c$, isolated by antisymmetrising in $\gamma \in \{\pm\frac{1}{4}, \pm\frac{1}{2}\}$ and Richardson extrapolation, is compared in Table 6.5 with

$$S = -\frac{mpq_1}{2} W(q_1) + \mathcal{L}\,\frac{(1+m+p)\,d_1}{p}, \qquad q_1 := \frac{m+1}{p}, \qquad \mathcal{L} := -mW(q)\pi\cot(\pi q), \tag{6.1}$$

per unit γ, with $d_1 = -(s-1)/2$. The first term is the response to the change of node density $\log|t'|$, a tilt $\propto z^{q_1}$ branched at the hard edge; the second is the leading pole term evaluated at the shifted index $n + (s-1)/(2p) = n + \delta_N$.

p	s	m	q_1	S fitted	(6.1)	role
2	2	1	1	-1.233702 $(1\cdot10^{-6})$	-1.233701	identification
2.5	2	1	0.8	-0.258693 $(2\cdot10^{-5})$	-0.258696	identification
3	2	1	0.667	$+0.371552$ $(1\cdot10^{-6})$	$+0.371551$	identification
2.2	2	1	0.909	-0.777016 $(7\cdot10^{-5})$	-0.776976	prediction
4	2	1	0.5	$+1.261975$ $(4\cdot10^{-7})$	$+1.261975$	prediction
3	2	2	1	-8.270517 $(3\cdot10^{-5})$	-8.270552	prediction
3	3	1	0.667	$+1.883052$ $(1\cdot10^{-6})$	$+1.883050$	prediction
2.5	3	1	0.8	$+0.659993$ $(2\cdot10^{-5})$	$+0.659993$	prediction
1.8	2	1	1.111	-1.847995 $(3\cdot10^{-4})$	-1.849477	see text
1.5	2	1	1.333	-3.39874 $(1.4\cdot10^{-2})$	-3.44204	see text
1.25	2	1	1.6	-6.54 (0.26)	-7.02	see text

TABLE 6.5. The subleading pole coefficient (script `spec_subpole`). At $p = 2$, $\mathcal{L} = 0$ and $S = -W(1) = -\pi^2/8$.

For $q_1 \le 1$ the formula holds to the fit precision in the eight rows; three of them were used to form it. For $q_1 > 1$ the exponents crowd near $p = 1$ and the fits depend on the nuisance pool: refits of the stored data over twenty pools (windows $n \ge 150$ to 600, three to seven nuisance exponents) give medians minus prediction of $-2\cdot10^{-3}$ ($p = 1.5$), $+3\cdot10^{-3}$ ($p = 1.8$) and $-9\cdot10^{-2}$ ($p = 2.5$, $m = 2$), with signs depending on the pool. The same decomposition, at $p = 2$, $m = 1$, gives a nonzero n^{-3} coefficient -3.70γ for this family; for Bessel zeros (§5.2), whose index correction at $p = 2$ is tied to the offset, the corresponding terms cancel, and the expansion is the pure index shift of Proposition 5.2.

6.5. **Drums.** Let λ run over the Dirichlet eigenvalues of a planar domain of area $\mathcal{A}$ and perimeter $\mathcal{P}$, rescaled to $\tilde\lambda = \mathcal{A}\lambda/4\pi$, so that the counting function is $\tilde\lambda - \beta\tilde\lambda^{1/2} + c_0 + (\text{fluctuations})$ with $\beta = \mathcal{P}/\sqrt{4\pi\mathcal{A}}$; the constant $c_0 = Z(0)$ carries corners and curvature ($\frac{1}{4}$ for rectangles, $\frac{1}{3}$ for

the equilateral triangle, $\frac{1}{6}$ for smooth simply connected domains). We take $p=1$, $s=2$, which is outside the range of the theorems. A *smoothed drum* is the node set defined by the counting function without fluctuations, $\mathcal{N}(\lambda_k) = k - \frac{1}{2} - c_0$. The perimeter is a pole of $Z(-w)$ at $q=-\frac{1}{2}$ with residue $\beta/2$, where $\cot(\pi q)=0$ and W has a pole; the limit $W(q)\pi\cot(\pi q) \to -\pi(4c)^{-1/2}$ of (3.3) gives the prediction $n(y_n - y_n^{\mathrm{lat}}) \supset -\frac{\pi\beta}{2\sqrt{2}} n^{1/2}$, which is the first term of Remark 3.6.

smoothed drum	β	fitted $n^{1/2}$ coefficient	$-\pi\beta/(2\sqrt{2})$
square	1.128379	-1.253317 $(1\cdot10^{-5})$	-1.253314
equilateral triangle	1.286074	-1.428473 $(1\cdot10^{-5})$	-1.428469
disk	1	-1.110722 $(7\cdot10^{-6})$	-1.110721
golden rectangle	1.161199	-1.289770 $(1\cdot10^{-5})$	-1.289767

TABLE 6.6. The perimeter term at $p=1$ (script `spec_corners`). The prediction, made before the run, was that no $n^{1/2}$ term occurs, which forgot the pole of W; the value in the last column is the $0\cdot\infty$ limit of the same formula, with no free parameter.

The corner term is isolated by comparing two smoothed drums with the same β (the square's) and $c_0 = \frac{1}{4}, \frac{1}{3}$: the n^0 coefficient of the difference is 0.0833487 $(2\cdot10^{-5})$, against $\Delta c_0 = \frac{1}{12}$, as the term $pZ(0)$ of the dictionary requires. The absolute n^0 coefficient has median $c_0 + \frac{1}{2} + \frac{\pi^2}{16}\beta^2$ over fits in all smoothed drums, to $2\cdot10^{-4}$, but single fits of it are unstable; the β^2 part is the second term of Remark 3.6, i.e. it is produced by solving the continuum soft-edge condition exactly rather than to first order. For the actual eigenvalues of the golden rectangle, the deviation from the smoothed drum has mean near 0 and a standard deviation growing like $n^{0.55}$: the counting fluctuations of the spectrum dominate everything below the perimeter term (§6.7).

6.6. **Anharmonic oscillators.** For the eigenvalues E_K $(K \ge 1)$ of $-\frac{\mathrm{d}^2}{\mathrm{d}q^2} + q^N$ on $\mathbb{R}$, $N \ge 4$ even, the semiclassical quantisation reads $K - \frac{1}{2} = AE^a + BE^{-a} + O(E^{-3a})$, $a = \frac{1}{2} + \frac{1}{N}$, with $A = \frac{1}{\pi}\int_{-1}^{1}\sqrt{1-t^N}\,\mathrm{d}t = \frac{2}{\pi N}B(\frac{1}{N},\frac{3}{2})$. With $p := 1/a = 2N/(N+2)$ and $\lambda_K := A^p E_K$,

$$\lambda_K^{1/p} = k' + \gamma/k' + O(k'^{-3}), \qquad k' = K - \tfrac{1}{2}, \qquad \gamma = -AB,$$

a Hurwitz lattice with $a = \frac{1}{2}$ plus an index correction with $m=1$, whose pole $q = 1/p$ has $W(q) = 1$ and $\cot(\pi/p) = -\tan(\pi/N) \ne 0$. (For Bessel zeros, and for linear potentials, the poles of $Z(-w)$ sit at half-integers, where the cotangent vanishes; here it does not.) The predictions are $d_1 = -\frac{s-1}{2} + \frac{p}{2}$ $(Z(0) = \zeta(0,\frac{1}{2}) = 0)$, no n^{-1-p} term (the heat-trace exponents of q^N contain no integer, so $Z(-1)=0$), and an n^{-2} coefficient equal to the Hurwitz-shifted lattice value plus the pole term $-\gamma\pi\cot(\pi/p) = AB\pi\cot(\pi/p)$.

The pole term is rational. If the second-order quantisation condition

$$\frac{1}{\pi}\int_{-q_+}^{q_+}\sqrt{E-V}\,\mathrm{d}q - \frac{1}{24\pi}\,\partial_E^2\int_{-q_+}^{q_+} V'(q)^2(E-V)^{-1/2}\,\mathrm{d}q = K - \tfrac{1}{2} + O(E^{-3a})$$

holds, then for $V = q^N$ the substitutions $q = E^{1/N}t$, $u = t^N$ give $B = -\frac{N}{12\pi}e(e-1)B(2-\frac{1}{N},\frac{1}{2})$ with $e = \frac{3}{2} - \frac{1}{N}$, and the reflection formulas collapse the product to

$$AB = -\frac{(N-1)\cot(\pi/N)}{6\pi(N+2)}, \qquad \text{hence} \qquad AB\,\pi\cot\frac{\pi}{p} = \frac{N-1}{6(N+2)},$$

since $\pi/p = \pi/2 + \pi/N$. The cotangent of the kernel cancels the $\cot(\pi/N)$ of the semiclassical coefficient. The values are $\frac{1}{12}, \frac{5}{48}, \frac{7}{60}, \frac{1}{8}, \frac{11}{84}$ for $N = 4, \dots, 12$; with B fitted from exact eigenvalues they are reproduced to between 10^{-10} and $7 \cdot 10^{-8}$ (script `spec_homog`); at $N = 4$ the fitted $B = -0.0476724854$ equals $-1/(12\pi A)$ to $4 \cdot 10^{-10}$.

N	d_1 fitted (predicted)	n^{-1-p} fitted	n^{-2} fitted	predicted	without pole term
4	$+0.16666667$ ($\frac{1}{6}$)	$1 \cdot 10^{-5}$ ($3 \cdot 10^{-3}$)	$+0.276010$ ($1 \cdot 10^{-5}$)	$+0.276042$	$+0.192708$
6	$+0.25000000$ ($\frac{1}{4}$)	$5 \cdot 10^{-5}$ ($2 \cdot 10^{-3}$)	$+0.315918$ ($3 \cdot 10^{-5}$)	$+0.315972$	$+0.211806$

TABLE 6.7. Oscillators $-\frac{\mathrm{d}^2}{\mathrm{d}q^2} + q^N$, $s = 2$, eigenvalues from a banded harmonic-oscillator basis converged to $3 \cdot 10^{-16}$ for $K \le 1200$, with the semiclassical series beyond (scripts `spec_quartic`, `spec_qN`). With the sign of the pole term reversed the predictions would be 0.109375 and 0.107639.

6.7. **Irregular spectra.** If the nodes deviate from a smooth node function by a fixed fraction of the spacing, the expansion is replaced by an irregular component. For iid jitter $\lambda_m = (m + \xi_m)^p$, ξ_m uniform on $(-0.3, 0.3)$, the residual of y_n after a smooth fit has root mean square of order $n^{-1/2}$ (for example $1.3 \cdot 10^{-2}$ on $[100, 200)$ and $4.2 \cdot 10^{-3}$ on $[800, 1600)$ at $p = 0.6$), and the fitted d_1 is meaningless. Jitter confined to nodes deep inside the band for all n considered is invisible (root mean square 10^{-9} to $3 \cdot 10^{-7}$), and jitter confined to the first 300 nodes stops producing noise once the soft-edge threshold $\lambda \approx n^p/(4c)$ has passed them (scripts `spec_jitter`, `spec_window`).

Single-node kernel. Moving node $k_0 \in \{100, 200, 400, 600\}$ by ± 0.05 and following $F(n; k_0) := [D(+h) - D(-h)]/(2h)$, $D = n(y_n - y_n^{\mathrm{lat}})$, in $\tau = (n - n_*)/n_*^{1/3}$, where n_* is the index at which the node crosses the edge, gives one curve for all k_0: outside the band ($\tau \ge 2$) it is the derivative of the per-node law $-p(1-u)^{-1/2}$ to 1–3%; near $\tau = 0$ it stays finite; inside the band it oscillates with a period of about ten nodes, and its sums over n nearly cancel (script `spec_edgekernel`).

Two ensembles. Replacing the first 3000 nodes of the lattice ($p = 0.6$, $s = 2$) by unfolded eigenphases of random unitary matrices or by a Poisson bridge gives the root mean square of D in Table 6.8. Both grow like $n^{1/2}$ (exponents 0.50 and 0.52), and the Poisson amplitude is larger by the stable factor 3.3–4.0. The noise is not a linear statistic realisation by realisation (correlations between -0.04 and 0.47 between D and its linearisation), but the linearisation has the same root mean square in every window (script `spec_noiselin`). The power spectrum of the linear kernel, measured with sinusoidal displacements $\epsilon \sin(2\pi k/\Lambda_0)$, scales as $n\kappa(\omega)$ with $\kappa(\omega) \propto \omega$ for small ω (script `spec_kernelfft`). The formula $\operatorname{Var} D \approx n \int_0^\pi \kappa(\omega) S_h(\omega)\, \mathrm{d}\omega/\pi$, with displacement spectrum $S_h = K(\omega)/(4 \sin^2(\omega/2))$, $K = 1$ (Poisson) and $K = \min(\omega/2\pi, 1)$ (unitary ensemble), predicts a ratio 3.95 and root mean squares 5.67, 8.02, 11.36 (unitary) and 22.5, 31.7, 44.9 (Poisson) on the last three windows, against the measured 5.38, 7.46, 11.56 and 20.8, 28.3, 41.8 for one seed.

7. Conjectures and open problems

The theorems of Sections 3–5 concern the lattice, apart from the potential theory of Section 3, which holds for general densities. This section states what they and the evidence of Section 6 suggest for other node sets, and what is missing for proofs.

n-window	unitary eigenphases	Poisson bridge	ratio
$[50, 100)$	2.67 ± 0.36	8.7 ± 2.5	3.3
$[100, 200)$	3.76 ± 0.34	15.0 ± 4.8	4.0
$[200, 400)$	5.38 ± 0.18	19.3 ± 3.4	3.6
$[400, 800)$	7.40 ± 0.33	26.3 ± 2.1	3.6
$[800, 1600)$	10.68 ± 0.50	39.8 ± 5.1	3.7

TABLE 6.8. Root mean square of $n(y_n - y_n^{\mathrm{lat}})$, mean and standard deviation over five seeds (scripts `spec_stats`, `spec_stats_summary`).

7.1. **The natural-index law.** Call Λ *regular* if $\lambda_m^{1/p}$ has an asymptotic expansion $m+a-1+\sum_{i\ge1}\gamma_i m^{-r_i}$ with exponents $0<r_1<r_2<\cdots\to\infty$, or if Λ differs from such a set in finitely many elements. Then Z_Λ is meromorphic, $Z_\Lambda(0)$ and $Z_\Lambda(-j)$ are defined by continuation, and $w\mapsto Z_\Lambda(-w)-\zeta(-pw,a)$ has poles only at points q produced by the corrections $\gamma_i m^{-r_i}$ (at $q=r_i/p$ to first order).

Conjecture 7.1 (natural-index law)**.** *Let Λ be regular and $N_\Lambda := n+\frac{\alpha}{2}-Z_\Lambda(0)$. Then $n^p b_n/c = (n/N_\Lambda)^p F$, where F has an asymptotic expansion in powers of $1/N_\Lambda$ and of the spectral monomials below, with*

(a) *no term N_Λ^{-1}, and the term $f_2 N_\Lambda^{-2}$, with f_2 as in (5.1);*
(b) *for $j\ge1$ the term $pW(j)Z_\Lambda(-j)N_\Lambda^{-1-jp}$;*
(c) *for every pole $q\neq -1/p$ of $Z_\Lambda(-w)-\zeta(-pw)$, with residue R_q, the term $pR_qW(q)\pi\cot(\pi q)N_\Lambda^{-1-pq}$, with the value $-(4c)^q\widetilde{L}(q)$ at $q\in-\frac12-\mathbb{N}_0$.*

Coefficients of coincident exponents add. The statement concerns the leading power of each datum in (b) and (c); it says nothing about the terms of order N_Λ^{-2-pq} (Remark 7.2).

For the lattice, (a) and (b) are Theorem 5.7 (with $Z_\Lambda(0)=-\frac12$, $W(j)p\zeta(-jp)=\varepsilon_j$), and (c) is empty. For Bessel zeros the whole expansion is Proposition 5.2. For Hurwitz lattices, synthetic corrections, smoothed drums and the oscillators q^N the evidence is in Section 6. In the form (b)–(c) the two kinds of spectral data are the residues of one integrand,

$$\frac{p}{n}\cdot\frac{1}{2\pi i}\int_{\uparrow} -W(w)\,\pi\cot(\pi w)\,Z_\Lambda(-w)\,n^{-wp}\,\mathrm{d}w,$$

at the integers (Theorem 4.2) and at the poles of Z_Λ (Proposition 3.5). The Weyl pole of Z_Λ is excluded: it is the leading term n of the soft-edge condition, not a correction. A moved Weyl pole, as for $\lambda_k=k^p+\kappa$, is included (§6.3). If a pole meets an integer, the two residues diverge separately and their sum produces a term $n^{-1-jp}\log n$; one weak test at $p=m=\frac12$ is consistent with the predicted coefficient.

Two consequences make the law a spectral dictionary. Through $Z_\Lambda(0)$, the $1/n$ coefficient $d_1=-p\alpha/2+pZ_\Lambda(0)$ reads off the index offset of the spectrum, such as the constant a of a Hurwitz lattice (§6.2) or the corner and curvature constant of a drum (§6.5). Through $Z_\Lambda(-j)$ and the poles, it reads off the coefficients of the heat trace of the operator whose spectrum is Λ, weighted by $W(q)\pi\cot(\pi q)$.

Remark 7.2 (subleading pole terms). Each pole also contributes at order n^{-2-pq}, one order below its leading term, and this contribution is not removed by $\cot(\pi q)=0$. For Bessel zeros at $p=2$ the McMahon correction has $q=\frac12$; its leading term vanishes, and its subleading term at n^{-3} cancels the zeta term $2pcZ_\Lambda(-1)$ exactly, which is why Proposition 5.2 is a pure

index shift; a pure Hurwitz lattice with the same offset does carry the zeta term at n^{-3}. For the synthetic family the subleading term is (6.1) when $q_1 \le 1$:

Conjecture 7.3. *For $\lambda_k = (k+\gamma k^{-m})^p$, $m>0$, $p\neq 1$, $q_1=(m+1)/p\le 1$, the part linear in γ of the n^{-2-m} coefficient of $n^p b_n/c$ is γS with S as in (6.1).*

Both terms of (6.1) are of the type that Conjecture 7.1 does not describe: a change of node density on the band, and an index shift of a first-order term. In Theorem 5.7 the same shift is visible as the factor $(n/N)^p$ in front of every layer. This suggests that every first-order term is to be evaluated at the natural index, plus a density term.

7.2. **The natural form to all orders.**

Conjecture 7.4. *For the lattice and $p\neq 1$, $F := (N/n)^p n^p b_n/c$ depends on n only through N^{-2} and the monomials $X_j := \varepsilon_j N^{-1-jp}$: $F = \sum_\beta c_\beta(N^{-2})\prod_j X_j^{\beta_j}$, with each c_β a power series.*

Theorem 5.7 proves the instances $\beta=0$ to N^{-6}, $\beta=e_j$ to relative order N^{-3} for every j (and N^{-4} for $j=1$), and $\beta=2e_1$ to relative order N^{-3}, with $c_{2e_1}(0)=(3p+1)/(2p)$. The numbers $(3p+1)/(2p)=1+(p+1)/(2p)$, like ε_j, do not depend on α. A closed form for c_0 is known only at $(p,\alpha)=(2,-\frac12)$, where it is $(1-x/4)^{-1/2}$ in $x=N^{-2}$; the hard-edge denominators $\prod_k((2k+1)p+2)$ of Remark 5.8(b) suggest a hypergeometric structure in $a'=\frac1p+\frac12$.

7.3. **Reflection and shift.** The exponent α enters the jump data of Lemma 5.3 in two ways: as the Bessel order ν in the coefficients $a_k(\nu), b_k(\nu)$, which are polynomials in ν^2, and as the Szegő exponent β in $\rho^2=\phi^{2(\beta+1)}$ at the hard edge and in the factor $\phi^{-2(\beta+1)}$ at the soft edge, where $\phi(z)=(z^{1/2}+(z-4c)^{1/2})/(2c^{1/2})$ is the exterior conformal map of $[0,4c]$. For the lattice $\nu=\beta=\alpha$. Keeping ν and β independent defines formal problems whose residue recursion (Section 5) gives formal series $b(n;\beta,\nu)$ in $1/n$.

Theorem 7.5 (reflection)**.** *As formal series, $b(n;\beta,\nu)=b(-n;-\beta-2,\nu)$; that is, $d_k(-\beta-2,\nu)=(-1)^k d_k(\beta,\nu)$ for all k.*

Proof. With $T=\frac{1}{\sqrt2}\left(\begin{smallmatrix}1 & 1\\ i & -i\end{smallmatrix}\right)$ we have $\sigma_2 T = T\sigma_3$, so $P_0^\infty = T\mathfrak{a}_\infty^{\sigma_3}T^{-1}$ and $P_1=-c\sigma_2$ commute with σ_2. The core matrices X_k and $\sigma_3\Xi_k\sigma_3$ lie in the span of I,σ_2 for even k and of $\sigma_3, i\sigma_1$ for odd k, so conjugation by σ_2 multiplies them by $(-1)^k$, while $\sigma_2 d^{\sigma_3}\sigma_2 = d^{-\sigma_3}$ for scalars d. Hence $\sigma_2\Delta_k(\beta)n^{-k}\sigma_2=\Delta_k(-\beta-2)(-n)^{-k}$, the recursion is covariant under conjugation by a constant matrix, and $(Q_1)_{12}(Q_1)_{21}$ is invariant under conjugation by σ_2. □

Lemma 7.6 (self-similar Buyarov–Rakhmanov identity)**.** *Let Φ denote the phase of the lattice equilibrium in the scaled frame, with hard-edge branch $2f^{1/2}$ and soft-edge branch $\tilde\varphi$. Then $(1+pz\partial_z)\Phi=-2\log\phi$ up to the additive constants of the branches; in particular $(1+pz\partial_z)(2f^{1/2})=2\operatorname{arsinh}(v/2)$ at $z=-v^2$ ($c=1$).*

Proof. At the hard edge $2f^{1/2}=\pi\sqrt{-z}\,\mathfrak{a}(z)$, and the series of $\mathfrak{a}$ has the factors $1/(a'+k)$ with $a'=\frac1p+\frac12$. Since $(1+pz\partial_z)[\sqrt{-z}\,z^k]=p(a'+k)\sqrt{-z}\,z^k$, these factors cancel term by term and leave the arcsine series. The soft-edge branch is the analytic continuation of the same phase. □

Under the self-similar scaling of the lattice problem, $1+pz\partial_z$ is the derivative in n of $n\Phi(xn^p)$; so Lemma 7.6 says that the n-flow of the phase is generated by $\log\phi$, which also generates the β-flow. Exact computer algebra (to order n^{-4} symbolically in (p,ν,β)) and 40-digit computation (to order n^{-8}, and for the zeta layer) support the following.

Conjecture 7.7 (shift)**.** *$b(n;\beta,\nu)\,n^p/c = (n/N)^p F_p(N;\nu^2)$ with $N = n + \frac{\beta+1}{2}$ and F_p independent of β; the same holds with the band-factor parameters included.*

Theorem 7.5 and Conjecture 7.7 together give, to all orders, the evenness of the integer layer in N (apply the reflection at $N' = n - \frac{\beta+1}{2}$) and in α (through ν^2), and they identify the lattice's natural index $n + \frac{\alpha}{2} + \frac{1}{2}$ as the Szegő shift. For varying weights e^{-nV}, evenness after a half-index shift is proved through the string equations [3]; an analogue of the string equation for the present formal problems would be a natural route to Conjecture 7.7.

Problem 7.8. *Find a second-order edge kernel: a function $g(w)$, analytic in w, with $g(j) = g_j$ for integers $j \geq 1$, such that the N^{-2} dressing of a pole term at q is $g(q)$. By Remark 5.8(d) it should combine a polynomial in $\theta = u\frac{\mathrm{d}}{\mathrm{d}u}$ acting on $(1-u)^{-1/2}$ with a multiple of $(1-u)^{1/2}$. The data of Table 6.5 and of drums are test cases.*

7.4. **What a proof needs.** The derivation of Conjecture 7.1(c) from the analysis of [10] needs that analysis for a node set other than the lattice, with three ingredients: the g-function of the constrained equilibrium of the n-dependent continuum node density (Theorem 3.2); the band factor given by the Cauchy integral of the jump of $\log(1 - e^{-2\pi i t_\kappa})$ for the index function $t_\kappa(z) = \kappa(n^p/z)$ (Proposition 4.4); and parametrix corrections whose first-order dependence on the perturbation is $O(n^{-2-m})$. Under these, the first-order change of $n^p b_n/c$ at order n^{-1-m} is the edge shift of (3.1) for the perturbed density, and Proposition 3.5 gives its value. For Hurwitz lattices the band factor is unchanged in form (its moments are Hurwitz zeta values), and what is missing is the index normalisation of the outer function, where the shift $-p\delta$ of d_1 must arise.

Remark 7.9 (a proved instance of the per-node law). Conjecture 7.1, read at leading order per node, says that one node with $u = 4c\lambda n^{-p} < 1$ outside the band contributes $-p(1-u)^{-1/2}/n$ to y_n when it is removed: this is the derivative of the soft-edge condition of Remark 3.3(c). For one atom k_0^{-p} of $\sigma_{p,s}$ this is proved, uniformly for $u_n = 4c(k_0/n)^p \leq 1-\eta$, with error $O(n^{-1-\epsilon})$, $\epsilon = \min(2p, 1)$ for $p < 1$ and $\epsilon = 1$ for $p \geq 1$, from the rank-one update of the Hankel matrices and a contraction argument for the minimal solution of the recurrence at the deleted node; the proof uses only the first-order expansions of a_n and b_n.

7.5. **Further problems.** (a) *The case $p = 1$.* Drums and the measures $\sum_k k^{-s}\delta_{1/k}$ have $p = 1$, where the two cotangents of Theorem 4.2 merge and [10] uses a different hard-edge analysis; Section 6.5 indicates that the dictionary persists. (b) *Higher layers.* The layers $\mu_1\mu_2$, μ_2^2, μ_1^3 control the coefficients below n^{-2-3p} and are not computed. (c) *Irregular spectra.* For spectra with displacements of the order of the spacing there is no deterministic expansion; Section 6.7 suggests that the variance of the irregular part is $n\int \kappa(\omega) S_h(\omega)\,\mathrm{d}\omega/\pi$ with a universal kernel spectrum κ. A closed form for κ, presumably from the Airy kernel of the soft edge, is open, as is a proof that rigidity at wavelengths of two to forty spacings sets the noise.

8. Relation to the literature

Constrained equilibrium. Existence, uniqueness and the structure (band, saturated region, balayage) of equilibrium problems with an upper constraint are classical [8, 5]. For conjugate-gradient convergence Beckermann and Kuijlaars derived endpoint equations of the same Cauchy-integral form as (3.1), with an external field and compactly supported constraints [2]; a formula for the case without external field is attributed to [1], which we have not been able to consult. We have not found the explicit density formula of Theorem 3.2, the criterion that $y\sigma(y)$ be decreasing, or the treatment of a constraint of infinite mass with a hard edge at

0. The fully saturated equilibrium of [14] ($y\sigma$ constant on a q-lattice) is the boundary case of hypothesis (H2).

Spectral data and Jacobi parameters. Orthogonal polynomials on reciprocal spectra, with asymptotics at fixed argument expressed through a regularised characteristic function, are studied in [12]. We know of no expansion of the recurrence coefficients themselves in terms of spectral zeta values at negative arguments, as in Theorem 4.2, and no reading of geometric spectral invariants (perimeter, corners) from Jacobi parameters. The Lommel family [13] is the classical exactly solvable case (Proposition 5.2).

Expansions in a shifted index. For varying weights e^{-nV} the recurrence coefficients have expansions in even powers of $1/n$ after a half-index shift; the odd terms are removed by a reindexing symmetry of the string equations [3], and for even weights the coefficients are generating functions of maps [6]. For Jacobi-type weights on $[-1,1]$ the first correction vanishes [7]. The present setting is a fixed discrete measure without a string equation: the shift involves the spectral constant $Z(0)$, and the evenness comes with the zeta layer and its own even dressing (Theorem 5.7, Conjecture 7.4). Theorem 7.5 plays the role of the reindexing symmetry of [3].

Appendix A. Reproducibility

All scripts are in the directory `research/G1/RH/spectral/checks` of the repository named in the data availability statement, next to their stored outputs (`.out`). They use the project's toolkit (quadruple-precision recurrence coefficients, cached under names that include a hash of the node array and of (p, s, K), so that a changed measure never reuses a cache), `mpmath` and `sympy`. The scripts that test a prediction end with assertions on it and print a line of the form "ALL n CHECKS PASSED" or "ALL IDENTITIES HOLD".

Exact computer algebra (Lemma 5.6). The outputs begin with the SHA-256 hashes of the scripts that produced them:

script	SHA-256
`m4_rh_exact.py`	`1e2904089fa3053cd6a7249dd4cd6149 2be3bb9c90b46c0932ff2507c9d9ffcc`
`m4_zeta_exact.py`	`3df62305bba0a4801a767c252eeee000 31e499f7c80a36946542fea6cd834041`
`m4_rh_exact2.py`	`143a308b8744bfccf3110c3a3f4a60a8 f192268b0f65f78b64bf8bd76c116521`
`m4_zeta_exact2.py`	`badda6a534fbf621c2bbef3eca6c7822 b26d13493a655a1655860449a8bab9f4`

The runs and what they assert, identically in the symbols:

output	configuration	content
`m4_rh_exact_k4.out`	$K = 4$	Lemma 5.6(a) to f_4
`m4_rh_exact_k6.out`	$K = 6$	Lemma 5.6(a): d_1, d_2, $f_1 = f_3 = f_5 = 0$, f_2, f_4, f_6
`m4_zeta_exact.out`	$K = 5$, $j = 1, \dots, 5$	Lemma 5.6(b) for integer j; $g_{1,4}$
`m4_zeta_exact2_A.out`	$K = 4$, j symbolic	Lemma 5.6(b) for all j
`m4_zeta_exact2_B.out`	$K = 5$, second order in μ_1	Lemma 5.6(c)

The first two runs also assert the coefficient identities $b_k = a_k + (k - \frac{1}{2})a_{k-1}$ (the termwise derivative of [4, (10.40.2)]) and the recursion for u_k, v_k of [4, (9.7.2)] for $k \le 8$; the symbolic run `m4_zeta_exact2_A` asserts the Taylor data of $Q_j/\binom{2j}{j}$ against the integer polynomials for $j \le 8$.

Floating-point scripts.

used in	scripts
Remark 3.7	`m1_theoremQ`, `m1_edge`
§4	`spec_kernel` (parts K0, K, P, Q), `m2_thmK_plt1`
§5.2	`spec_lommel`
Remark 5.8(b)	`m4_rh4`, `m4_f8_denominator`
§6.1	`m4_natural_index`, `m4_f4_lattice`, `m4_zeta_lattice`
§6.2	`spec_hur`, `spec_hur_d2`, `spec_shift_hi`, `spec_hur_small`
§6.3	`spec_pole`, `spec_gamma`, `spec_kappa`, `spec_kappa_fit`, `spec_mobius`, `spec_mobius_pred`, `spec_resonance`
§6.4	`spec_subpole`
§6.5	`spec_corners`
§6.6	`spec_quartic`, `spec_qN`, `spec_homog`
§6.7	`spec_jitter`, `spec_window`, `spec_edgekernel`, `spec_stats`, `spec_stats_summary`, `spec_noiselin`, `spec_kernelfft`

The shell script `run_all.sh` in the same directory regenerates the outputs from scratch in dependency order (readers of other scripts' caches last), including the five exact runs (about 35 minutes) and the coefficient computations (one to two and a half minutes per measure with $n \le 1600$ on two cores). The scripts `spec_quartic`, `spec_qN`, `spec_corners`, `spec_edgekernel`, `spec_noiselin` and `spec_kernelfft` are not in `run_all.sh` and are run individually.

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Data and code availability

The scripts that produced the exact identities of Section 5 and the numbers of Section 6, with their outputs, are available in the repository https://github.com/speagleg/jacobi-asymptotics; Appendix A lists them.

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